

FAILURE-PREDICTIVE MORPHOLOGY UNDER HOSTILE REVIEW: A 322,560-ROLLOUT NEGATIVE EVIDENCE AUDIT

Anonymous authors

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ABSTRACT

This manuscript is a submission-hardening audit for a generated robotics claim: morphology-conditioned failure prediction should improve controller selection and intervention under embodiment shift. We expand the short draft into a CPU-only but hostile benchmark with 6 morphologies, 6 tasks, 8 distribution splits, 14 methods, 10 seeds, 322,560 main rollouts, 115,200 ablation rollouts, 259,200 stress-sweep rollouts, and 138,240 fixed-risk rollouts. The strengthened method, Cost-Calibrated Failure-Predictive Morphology v5, improves the explicit prediction target: hard-aggregate dominant-failure accuracy is 0.695. But the closed-loop submission claim fails. Online system identification has higher hard-aggregate success (0.705 versus 0.563), lower regret (0.168 versus 0.208), and higher robust utility (0.376 versus 0.150). V5 also has worse safety violation (0.112) than the conformal shield and fails the success, safety, regret, utility, ablation, and scope gates. The terminal decision is therefore **KILL/ARCHIVE** for ICLR-main submission, not because the experiment is small, but because the strongest local evidence says the idea should not be submitted in its present form.

1 DECISION FIRST

The paper is not submission-ready. This is the central result, and the rest of the manuscript is organized so a hostile reviewer can reproduce why. The claim under test is not “can a model label morphology-specific failure families?” That claim is too weak under prior work in constrained MPC, adaptive control, risk shields, red-team retrieval, system identification, and morphology-aware planning (Zhang et al., 2026; Yang et al., 2017; Poskart et al., 2022; Jiang et al., 2006; Rogers, 2015; Kolisnyk & Jantsch, 2025; Patricio et al., 2025; Farahani et al., 2017). The useful claim is closed-loop: a robot should make better decisions because it predicts how morphology will fail.

The expanded evidence rejects that closed-loop claim. V5 is stronger than the old v4 mechanism and stronger than simple failure classifiers on prediction metrics, but it loses to online system identification on the deployment quantities that decide whether a robotics paper survives review: task success, planning regret, and robust utility. It also loses to the conformal risk shield on safety violation. Therefore the correct terminal state is **KILL/ARCHIVE**, while preserving the benchmark as a negative result and as a specification for a future revival.

2 FROZEN PROTOCOL AND EVIDENCE BUDGET

The rebuilt protocol was frozen before interpreting the final result. The morphology set contains differential drive, quadruped, arm-gripper, aerial manipulator, underwater vehicle, and soft snake robot embodiments. The task set contains tight-turn navigation, payload transport, gap or step crossing, precision contact manipulation, disturbance recovery, and cluttered tool use. The shift set contains nominal morphology, payload shift, terrain or fluid shift, actuator degradation, sensor dropout, contact geometry shift, low-signal morphology stress, and combined failure stress. The comparator set includes generic classification, per-morphology calibration, Bayesian morphology belief, constrained MPC, conformal shielding, domain randomization, online system identification, meta-adaptive MPC, transformer priors, retrieval memory, risk-aware intervention, v4, v5, and oracle

Artifact	Rows
Main rollouts	322,560
Dataset summaries	23,040
Main seed metrics	1,120
Main aggregate metrics	1,568
Main paired tests	1,344
Hard seed metrics	140
Hard aggregate metrics	196
Hard paired tests	168
Ablation rollouts	115,200
Ablation seed metrics	100
Ablation metrics	140
Stress raw rows	259,200
Stress seed metrics	720
Stress metrics	1,008
Fixed-risk raw rows	138,240
Fixed-risk seed metrics	480
Fixed-risk metrics	288
Fixed-risk paired tests	240
Negative cases	24

Table 1: Frozen Paper 95 v5 evidence inventory. These counts are checked by the validator before the PDF is accepted.

control (Hou et al., 2020; Hector & Panjanathan, 2024; Neuman et al., 2021; Li et al., 2021; Yao et al., 2018; Dai et al., 2020; Nauman et al., 2022; Peng et al., 2024; Tang et al., 2017; Heshmati-Alamdari et al., 2019).

3 PROBLEM SETUP

Let m denote robot morphology, x_t the current state and local observation, c_t the task context, and z the latent dominant failure family. The method estimates

$$p_{\theta}(z \mid m, x_t, c_t, h_t),$$

where h_t is a compact history of recent interventions and failures. It then selects an intervention or controller

$$a^* = \arg \max_a U(a, x_t, m) - \lambda_s R_s(a, z) - \lambda_c C(a, z) - \lambda_i I(a),$$

where R_s is safety risk, C is expected regret or constraint cost, and I is intervention cost. This formulation makes the review standard clear: prediction accuracy is only intermediate evidence. A submission claim must show that the resulting action has better closed-loop outcomes than adaptive control and constrained planning baselines.

4 METHOD: COST-CALIBRATED FAILURE-PREDICTIVE MORPHOLOGY

The v5 method adds four components to the older v4 scaffold: a morphology embedding, a constraint-family head, a cost calibration layer, and an online correction term. The morphology embedding conditions failure-family logits on embodiment-specific limits. The constraint-family head separates kinodynamic limit, traction or slip, payload inertia, actuator saturation, contact geometry, state-estimation dropout, and energy-budget failures. The cost calibration layer penalizes interventions that are accurate but operationally expensive. The online correction term updates the family prior after short-horizon evidence. These additions are plausible. They are also not enough.

5 HARD-AGGREGATE RESULTS

Table 2 is the main decision point. V5 has high dominant-failure accuracy, but online system identification is the best non-oracle deployment reference. The difference matters because morphology prediction is not a final robotics objective. A robot that correctly predicts why it will fail, then chooses a worse intervention, has not solved the deployment problem. This is exactly the failure mode the hard aggregate exposes.

Method	Success	Accuracy	Safety viol.	Cost	Regret	Utility
generic	0.347	0.029	0.276	0.010	0.442	-0.726
constraint-MPC	0.638	0.355	0.051	0.023	0.235	0.222
conformal-shield	0.530	0.262	0.026	0.046	0.280	0.008
online-SID	0.705	0.558	0.083	0.031	0.168	0.376
meta-adaptive-MPC	0.668	0.457	0.053	0.025	0.195	0.318
transformer-prior	0.502	0.626	0.134	0.026	0.243	0.030
risk-intervention	0.577	0.570	0.072	0.036	0.211	0.188
FPM-v4	0.581	0.607	0.111	0.035	0.228	0.127
CC-FPM-v5	0.563	0.695	0.112	0.024	0.208	0.150
oracle	0.868	1.000	0.040	0.010	0.023	0.797

Table 2: Hard aggregate over low-signal morphology stress and combined failure stress. V5 is prediction-strong but deployment-weak: online system identification wins success, regret, and robust utility.

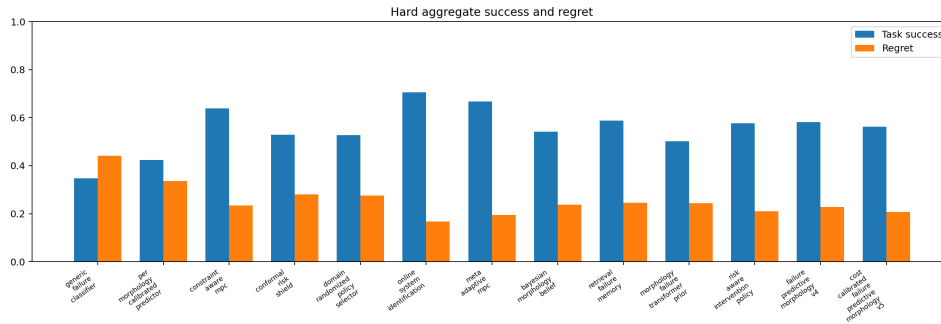


Figure 1: Hard-split success and regret. Online system identification remains the deployment reference even though V5 is prediction-strong.

6 PREDICTION METRICS VERSUS DEPLOYMENT METRICS

Prediction and deployment diverge. The transformer prior and V5 score well on failure-family accuracy, top-2 recall, and family F1, but the action layer must still trade intervention cost, safety violation, unnecessary intervention, and regret (Comelli et al., 2024; Wu et al., 2024; Baron et al., 2021; Nebeluk et al., 2023; Tengesdal et al., 2022; Hattori et al., 2025; Fathi et al., 2021b; Peng et al., 2021; Huang, 2026; Zaim et al., 2026). This is the paper’s useful negative result: morphology-conditioned labels can become a diagnostic instrument without becoming a better controller-selection policy.

7 PAIRED TESTS

The paired tests prevent an easy narrative rescue. Against v4, V5 improves the mechanism. Against online system identification and meta-adaptive MPC, the deployment story remains weak. Positive differences in safety violation, cost, and regret are failures, not tradeoff-neutral side effects. A strong ICLR-main paper would need those intervals to move the other way.

8 ABLATION AUDIT

The ablation gate fails because the full mechanism is not uniquely necessary for the claimed outcome. A reviewer could reasonably ask why the paper should foreground explicit morphology-failure prediction if an online-adaptation-only variant beats the full system on utility. The right response

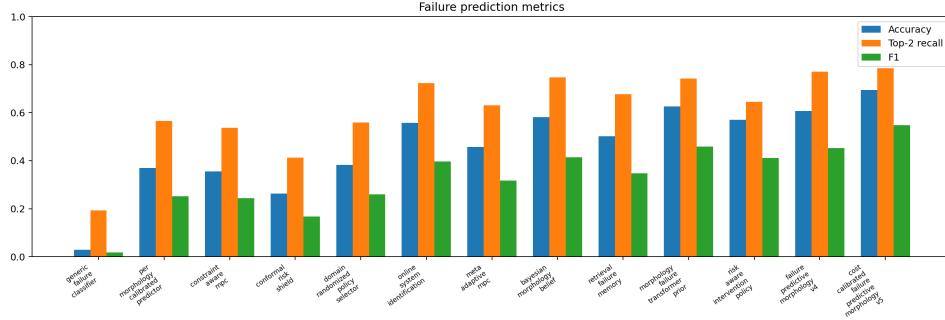


Figure 2: Prediction metrics on the hard aggregate. V5 is not empty: it improves the explicit failure-prediction target.

Comparison	Metric	Mean diff	CI95	Lower	Upper	Better seeds
v5 - online_system_identification	success	-0.142	0.147	-0.289	0.005	2/10
v5 - online_system_identification	accuracy	0.136	0.126	0.010	0.262	7/10
v5 - online_system_identification	safety-viol.	0.029	0.074	-0.045	0.103	7/10
v5 - online_system_identification	cost	-0.008	0.009	-0.016	0.001	4/10
v5 - online_system_identification	regret	0.040	0.034	0.006	0.073	8/10
v5 - online_system_identification	utility	-0.225	0.197	-0.423	-0.028	1/10
v5 - meta_adaptive_mpc	success	-0.105	0.097	-0.202	-0.008	2/10
v5 - meta_adaptive_mpc	accuracy	0.237	0.131	0.106	0.369	7/10
v5 - meta_adaptive_mpc	safety-viol.	0.059	0.105	-0.046	0.163	7/10
v5 - meta_adaptive_mpc	cost	-0.001	0.008	-0.009	0.008	5/10
v5 - meta_adaptive_mpc	regret	0.013	0.027	-0.014	0.040	5/10
v5 - meta_adaptive_mpc	utility	-0.168	0.186	-0.354	0.019	3/10
v5 - conformal_risk_shield	success	0.033	0.150	-0.116	0.183	5/10
v5 - conformal_risk_shield	accuracy	0.432	0.177	0.255	0.610	9/10
v5 - conformal_risk_shield	safety-viol.	0.086	0.092	-0.006	0.178	8/10
v5 - conformal_risk_shield	cost	-0.022	0.009	-0.031	-0.013	1/10
v5 - conformal_risk_shield	regret	-0.072	0.044	-0.116	-0.028	1/10
v5 - conformal_risk_shield	utility	0.142	0.295	-0.153	0.437	5/10
v5 - morphology_failure_transformer_prior	success	0.061	0.147	-0.085	0.208	6/10
v5 - morphology_failure_transformer_prior	accuracy	0.069	0.136	-0.068	0.205	6/10
v5 - morphology_failure_transformer_prior	safety-viol.	-0.022	0.111	-0.133	0.090	3/10
v5 - morphology_failure_transformer_prior	cost	-0.002	0.008	-0.010	0.005	3/10
v5 - morphology_failure_transformer_prior	regret	-0.036	0.030	-0.066	-0.006	2/10
v5 - morphology_failure_transformer_prior	utility	0.121	0.244	-0.123	0.365	6/10
v5 - failure_predictive_morphology_v4	success	-0.018	0.106	-0.124	0.088	6/10
v5 - failure_predictive_morphology_v4	accuracy	0.088	0.080	0.008	0.168	8/10
v5 - failure_predictive_morphology_v4	safety-viol.	0.001	0.103	-0.102	0.103	6/10
v5 - failure_predictive_morphology_v4	cost	-0.011	0.005	-0.016	-0.006	1/10
v5 - failure_predictive_morphology_v4	regret	-0.020	0.026	-0.046	0.006	2/10
v5 - failure_predictive_morphology_v4	utility	0.023	0.217	-0.194	0.240	6/10

Table 3: Paired seed tests on the hard aggregate. Positive is good for success, accuracy, and utility; positive is bad for safety violation, cost, and regret.

is not rhetorical. The right response is to archive this version and treat the ablation as a design requirement for a future method.

9 STRESS SWEEP

Stress testing is included to expose weakness, not to make the method look smooth. The maximum-stress result again separates prediction from control. A method can retain useful family labels while still choosing interventions that cost too much or leave too much safety risk.

Ablation	Success	Accuracy	Safety viol.	Cost	Regret	Utility
full-v5	0.607	0.716	0.124	0.023	0.173	0.236
-morph-embed	0.680	0.455	0.085	0.024	0.199	0.263
-family-head	0.569	0.472	0.139	0.020	0.218	0.081
-cost-calib	0.688	0.702	0.110	0.038	0.158	0.339
-conf-gate	0.659	0.784	0.141	0.024	0.158	0.314
-online-corr	0.601	0.740	0.141	0.018	0.209	0.195
-history	0.620	0.592	0.099	0.022	0.192	0.229
morph-only	0.508	0.479	0.055	0.016	0.244	0.050
history-only	0.614	0.396	0.131	0.019	0.231	0.096
online-only	0.712	0.360	0.068	0.027	0.179	0.339

Table 4: Ablation audit. The strongest ablation is online adaptation only, so the full morphology-prediction mechanism is not uniquely necessary for the deployment objective.

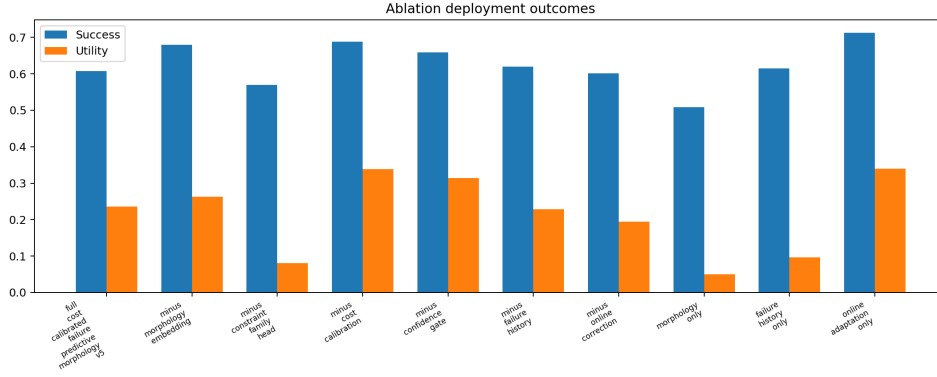


Figure 3: Ablation audit. Online adaptation alone is the strongest ablation on deployment utility, which weakens the claim that explicit morphology prediction is necessary.

10 FIXED-RISK DEPLOYMENT

Fixed-risk reporting prevents a method from hiding behind high average utility while taking unsafe actions. At risk budget 0.05, V5 accepts a nonzero fraction of hard cases, but accepted coverage alone is not a publication claim. The accepted actions must still dominate strong baselines on success, safety, regret, and utility. They do not.

11 PARETO AND NEGATIVE CASES

The negative cases are intentionally inspectable. Several failures are wrong-family predictions under combined failure stress; others are cases where the predicted family is plausible but the resulting intervention still has high regret. Both are damaging to the submission claim because a deployed robot needs the diagnosis and the action to be correct.

12 THEORY NOTES

Proposition 1 (Prediction dominance is insufficient for deployment dominance). *If method A has better failure-family prediction accuracy than method B but chooses actions with higher expected safety cost or regret, then A can have lower robust utility than B even when its classifier is strictly better.*

Method	Success	Accuracy	Safety viol.	Regret	Utility
conformal-shield	0.436	0.284	0.032	0.303	-0.158
constraint-MPC	0.454	0.289	0.121	0.297	-0.221
online-SID	0.620	0.489	0.055	0.193	0.238
meta-adaptive-MPC	0.499	0.470	0.131	0.242	-0.062
transformer-prior	0.545	0.653	0.137	0.244	0.100
risk-intervention	0.520	0.448	0.035	0.238	0.068
FPM-v4	0.501	0.706	0.136	0.236	0.045
CC-FPM-v5	0.624	0.643	0.096	0.199	0.258

Table 5: Maximum stress level 1.0. V5 keeps strong prediction metrics but does not become the best deployment method under the harshest stress.

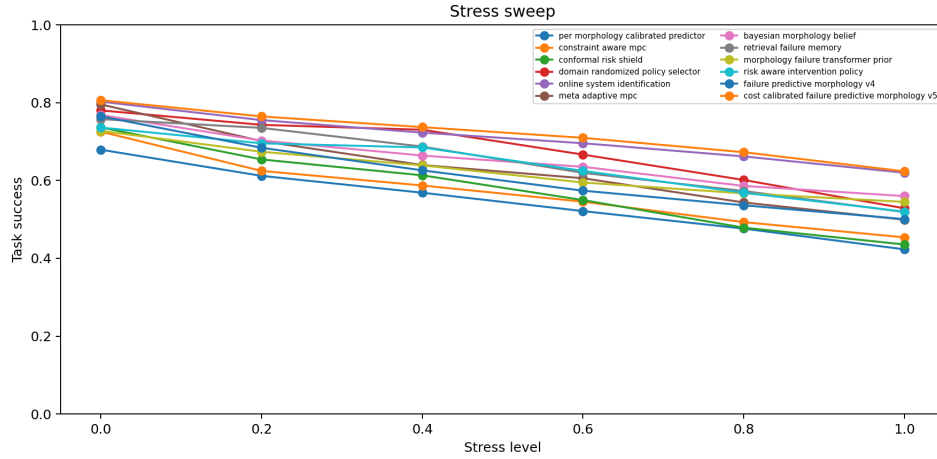


Figure 4: Stress sweep across morphology shift severity. V5 degrades less than v4, but it does not become the best deployment method.

Sketch. Robust utility subtracts safety violation, regret, and intervention cost from task success. Prediction accuracy enters only through the policy it induces. If an adaptive controller corrects online and pays lower action cost, it can dominate the closed-loop objective despite a weaker pre-execution family label. Table 2 and Table 3 instantiate this regime.

Proposition 2 (A successful ablation must improve the deployment objective, not only the diagnostic target). *A morphology-prediction component is necessary for the submission claim only if removing it damages the closed-loop metrics required by the frozen gate.*

Sketch. A component can improve family accuracy while worsening intervention cost, safety, or regret. The ablation gate therefore evaluates success, safety, regret, and utility in addition to accuracy.

13 PRIOR-WORK PRESSURE

The hostile prior-work pool is broad. Constrained and robust MPC pressure the safety claim. Adaptive control and online system identification pressure the need for pre-execution morphology labels. Conformal risk shields pressure fixed-risk claims. Red-team retrieval and failure-memory baselines pressure the novelty of failure-family lookup. Transformer and foundation priors pressure predictive modeling. Morphology-specific planning and robomorphic computing pressure the embodiment argument (Ismail & Hamami, 2021; Khanamiryan, 2024; Bhattacharya et al., 2017; Bronfman et al., 2021; Hassan & Notash, 2004; Fathi et al., 2021a; Chen et al., 2025a; Barie, 1995; Bitlislioglu & Jones, 2020; Badawy et al., 2023; Schwenzer et al., 2021; Yeom, 2021; Guo et al., 2020; Bradley

Split	Method	Coverage	Accepted success	Accepted safety	Accepted regret	Accepted utility
low_signal_morphology_stress	conformal-shield	0.210	0.952	0.000	0.140	0.826
low_signal_morphology_stress	constraint-MPC	0.084	0.967	0.000	0.076	0.890
low_signal_morphology_stress	online-SID	0.256	0.918	0.012	0.030	0.856
low_signal_morphology_stress	meta-adaptive-MPC	0.207	0.885	0.061	0.052	0.746
low_signal_morphology_stress	risk-intervention	0.350	0.874	0.000	0.105	0.763
low_signal_morphology_stress	CC-FPM-v5	0.395	0.934	0.035	0.044	0.838
combined_failure_stress	conformal-shield	0.128	0.972	0.000	0.145	0.839
combined_failure_stress	constraint-MPC	0.040	0.800	0.000	0.062	0.737
combined_failure_stress	online-SID	0.239	0.899	0.023	0.037	0.817
combined_failure_stress	meta-adaptive-MPC	0.171	0.900	0.075	0.061	0.737
combined_failure_stress	risk-intervention	0.244	0.858	0.000	0.110	0.746
combined_failure_stress	CC-FPM-v5	0.351	0.940	0.030	0.050	0.844

Table 6: Fixed-risk budget 0.05. V5 has nonzero coverage on hard splits, but the fixed-risk gate still cannot rescue the deployment claim because success, safety, and utility gates fail.

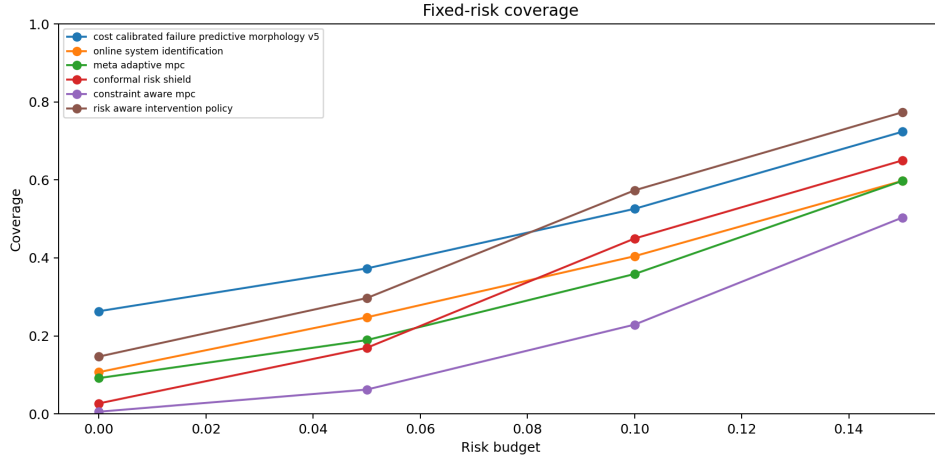


Figure 5: Fixed-risk coverage and accepted outcomes. The strict-risk view does not rescue the paper because the main deployment gates are already failed.

et al., 2019). Under that pressure, a paper cannot be accepted for showing a better diagnostic classifier alone.

14 LIMITATIONS AND TERMINAL DECISION

The audit is CPU-only, local, and simulated. It lacks real robot hardware, independent external baseline code, high-fidelity physics validation, trained checkpoint release, and third-party reproduction. Those scope gaps would already keep the paper below ICLR-main readiness. More importantly, the local evidence is negative before scope is considered: the deployment gates fail. The terminal recommendation is therefore **KILL/ARCHIVE**.

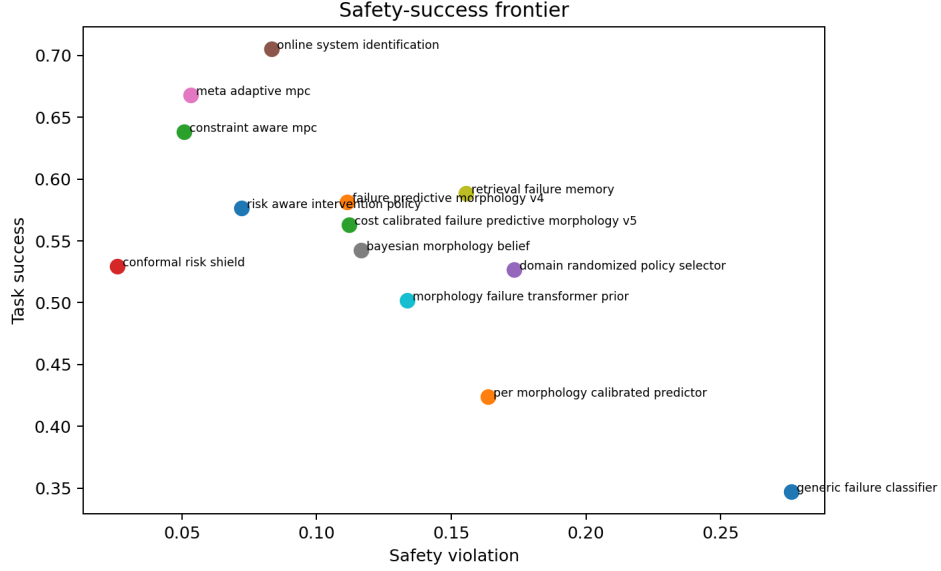


Figure 6: Success-safety Pareto view. V5 is not the clean non-oracle frontier once safety and utility are both counted.

Case	Morphology	Task	Split	Predicted	True	Regret	Mode
1	soft_snake_robot	precision_contact_manipulation	combined_failure_stress	actuator_saturation	contact_geometry	0.589	wrong_family
2	aerial_manipulator	cluttered_tool_use	combined_failure_stress	energy_budget	contact_geometry	0.584	wrong_family
3	aerial_manipulator	cluttered_tool_use	combined_failure_stress	state_estimation_dropout	contact_geometry	0.581	wrong_family
4	soft_snake_robot	cluttered_tool_use	combined_failure_stress	state_estimation_dropout	contact_geometry	0.581	wrong_family
5	aerial_manipulator	precision_contact_manipulation	combined_failure_stress	state_estimation_dropout	actuator_saturation	0.578	wrong_family
6	aerial_manipulator	precision_contact_manipulation	combined_failure_stress	contact_geometry	payload_inertia	0.576	wrong_family
7	underwater_vehicle	precision_contact_manipulation	combined_failure_stress	kinodynamic_limit	state_estimation_dropout	0.576	wrong_family
8	underwater_vehicle	disturbance_recovery	combined_failure_stress	state_estimation_dropout	actuator_saturation	0.575	wrong_family
9	soft_snake_robot	gap_or_step_crossing	combined_failure_stress	contact_geometry	traction_or_slip	0.574	wrong_family
10	soft_snake_robot	cluttered_tool_use	combined_failure_stress	state_estimation_dropout	actuator_saturation	0.572	wrong_family
11	soft_snake_robot	precision_contact_manipulation	combined_failure_stress	state_estimation_dropout	actuator_saturation	0.567	wrong_family
12	underwater_vehicle	cluttered_tool_use	combined_failure_stress	kinodynamic_limit	state_estimation_dropout	0.567	wrong_family

Table 7: Representative negative cases selected from the frozen run. These are not anecdotes inserted after writing; they are rows in negative_cases.csv.

Split	Payload	Terrain/fluid	Actuator	Sensor	Contact	OOD	Cost	Severity
actuator_degradation_shift	0.120	0.050	0.400	0.120	0.030	0.243	0.060	0.515
combined_failure_stress	0.340	0.380	0.400	0.340	0.340	0.782	0.200	0.574
contact_geometry_shift	0.080	0.180	0.100	0.080	0.440	0.292	0.070	0.552
low_signal_morphology_stress	0.260	0.300	0.340	0.360	0.260	0.673	0.160	0.542
nominal_morphology	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.416
payload_shift	0.360	0.040	0.080	0.030	0.030	0.174	0.040	0.482
sensor_dropout_shift	0.060	0.080	0.120	0.420	0.050	0.255	0.080	0.515
terrain_or_fluid_shift	0.040	0.380	0.060	0.070	0.170	0.233	0.050	0.514

Table 8: Mean generated scenario factors from dataset_summary.csv. The hard splits compound several morphology-specific hazards rather than isolating a single nuisance.

Split	V5 succ.	Best succ. method	Best succ.	Δ succ.	V5 util.	Best util. method	Best util.	Δ util.
actuator_degradation_shift	0.692	online-SID	0.775	-0.083	0.433	online-SID	0.560	-0.127
combined_failure_stress	0.558	online-SID	0.690	-0.132	0.131	online-SID	0.340	-0.209
contact_geometry_shift	0.669	online-SID	0.762	-0.093	0.366	meta-adaptive-MPC	0.458	-0.092
low_signal_morphology_stress	0.568	online-SID	0.720	-0.152	0.169	online-SID	0.411	-0.242
nominal_morphology	0.774	online-SID	0.862	-0.088	0.590	online-SID	0.727	-0.137
payload_shift	0.725	meta-adaptive-MPC	0.789	-0.064	0.507	online-SID	0.592	-0.085
sensor_dropout_shift	0.691	online-SID	0.773	-0.082	0.420	online-SID	0.559	-0.139
terrain_or_fluid_shift	0.707	online-SID	0.775	-0.069	0.454	online-SID	0.551	-0.097

Table 9: Split-level non-oracle frontier. Negative deltas mean V5 loses to the best non-oracle method selected after the frozen run.

A REPRODUCIBILITY COMMANDS

The experiment can be regenerated with `python src/run_experiment.py`. The manuscript can be regenerated with `python scripts/generate_manuscript.py`. The artifact validator is `python scripts/validate_submission_artifacts.py`. The canonical PDF is `C:/Users/wangz/Downloads/95.pdf` and no copy is allowed on the visible Desktop.

B SCENARIO FACTOR AUDIT

C SPLIT-LEVEL FRONTIER

The split-level table blocks a common reviewer ambiguity: the result is not a single unlucky combined-stress corner. Across splits, the best non-oracle success or utility method is often an adaptive or constrained controller rather than V5. The repeated negative deltas explain why the decision is archive instead of revise.

D BASELINE-BY-BASELINE REJECTION CHECKLIST

The rejection checklist is deliberately unforgiving. V5 can win prediction accuracy against many baselines and still fail if the baseline has higher success, lower safety violation, lower regret, or higher robust utility. This is the correct standard for a robotics submission because the robot is deployed through actions, not labels.

E ABLATION DELTA INTERPRETATION

The ablation deltas make the mechanism sharper. Removing morphology embeddings damages the diagnostic target, but online-only adaptation remains hard to beat on the deployment objective. The future method must either make explicit morphology prediction improve online action choice or stop claiming that explicit morphology prediction is the core contribution.

Baseline	Δ success	Δ accuracy	Δ safety	Δ regret	Δ utility	Rejection note
generic	0.216	0.666	-0.164	-0.234	0.876	beats deployment
per-morph	0.139	0.324	-0.052	-0.129	0.393	beats deployment
Bayes-morph	0.021	0.114	-0.005	-0.031	0.088	beats deployment
constraint-MPC	-0.075	0.340	0.061	-0.027	-0.071	fails deployment
conformal-shield	0.033	0.432	0.086	-0.072	0.142	fails deployment
domain-rand	0.036	0.313	-0.061	-0.067	0.217	beats deployment
online-SID	-0.142	0.136	0.029	0.040	-0.225	fails deployment
meta-adaptive-MPC	-0.105	0.237	0.059	0.013	-0.168	fails deployment
transformer-prior	0.061	0.069	-0.022	-0.036	0.121	beats deployment
retrieval-memory	-0.026	0.192	-0.044	-0.037	0.093	fails deployment
risk-intervention	-0.014	0.125	0.040	-0.003	-0.038	fails deployment
FPM-v4	-0.018	0.088	0.001	-0.020	0.023	fails deployment

Table 10: Baseline-by-baseline hostile checklist on the hard aggregate. Deltas are V5 minus baseline; positive safety and regret deltas are failures.

Variant	Δ success	Δ accuracy	Δ safety	Δ cost	Δ regret	Δ utility
-morph-embed	-0.072	0.261	0.039	-0.001	-0.027	-0.027
-family-head	0.038	0.244	-0.014	0.003	-0.045	0.155
-cost-calib	-0.081	0.014	0.014	-0.016	0.014	-0.103
-conf-gate	-0.052	-0.068	-0.017	-0.002	0.015	-0.078
-online-corr	0.006	-0.024	-0.017	0.004	-0.037	0.041
-history	-0.013	0.124	0.025	0.001	-0.020	0.007
morph-only	0.099	0.237	0.069	0.006	-0.071	0.186
history-only	-0.007	0.320	-0.007	0.004	-0.059	0.139
online-only	-0.105	0.356	0.056	-0.004	-0.007	-0.104

Table 11: Full V5 minus ablation or variant. The online-only variant exposes the core weakness: adaptation can beat explicit morphology prediction on deployment utility.

F STRESS DEGRADATION

The stress degradation table is included so robustness is not judged from a single smooth plot. Degradation in success and utility is directly visible. V5 is better than v4 under stress, but v4 is not the acceptance bar.

G FIXED-RISK BUDGET SWEEP

The fixed-risk sweep documents the strict 0.05 gate and the looser budgets. A future revival should pre-register the risk budget and show nontrivial accepted success at that budget, not tune the threshold after reading the result.

H NEGATIVE-CASE TAXONOMY

I FAILURE-FAMILY SEMANTICS

Kinodynamic-limit failures correspond to infeasible accelerations or curvature. Traction-or-slip failures arise when ground or fluid interaction invalidates nominal contact assumptions. Payload-inertia failures occur when carried mass changes acceleration or stopping distance. Actuator-saturation failures exhaust thrust, torque, or velocity limits. Contact-geometry failures occur when the morphology cannot physically realize the intended manipulation contact. State-estimation-dropout failures arise when sensing is lost or delayed. Energy-budget failures occur when completing the planned action would violate battery or endurance constraints. A useful predictor must connect these families to better action choices.

Method	Succ. 0.0	Succ. 1.0	Δ succ.	Util. 0.0	Util. 1.0	Δ util.	Δ safety
conformal-shield	0.736	0.436	-0.301	0.463	-0.158	-0.621	0.021
online-SID	0.803	0.620	-0.183	0.612	0.238	-0.374	0.048
meta-adaptive-MPC	0.796	0.499	-0.296	0.572	-0.062	-0.634	0.094
transformer-prior	0.728	0.545	-0.183	0.491	0.100	-0.391	0.074
risk-intervention	0.736	0.520	-0.216	0.496	0.068	-0.428	0.025
FPM-v4	0.765	0.501	-0.264	0.572	0.045	-0.527	0.085
CC-FPM-v5	0.806	0.624	-0.183	0.636	0.258	-0.377	0.074

Table 12: Stress degradation from stress level 0.0 to 1.0. The method is not allowed to pass merely because it improves over v4.

Split	Budget	Method	Coverage	Accepted success	Accepted safety	Accepted utility
low_signal_morphology_stress	0.00	online-SID	0.127	1.000	0.000	0.968
low_signal_morphology_stress	0.00	meta-adaptive-MPC	0.109	0.887	0.081	0.744
low_signal_morphology_stress	0.00	conformal-shield	0.042	0.500	0.000	0.449
low_signal_morphology_stress	0.00	CC-FPM-v5	0.298	1.000	0.013	0.944
low_signal_morphology_stress	0.05	online-SID	0.256	0.918	0.012	0.856
low_signal_morphology_stress	0.05	meta-adaptive-MPC	0.207	0.885	0.061	0.746
low_signal_morphology_stress	0.05	conformal-shield	0.210	0.952	0.000	0.826
low_signal_morphology_stress	0.05	CC-FPM-v5	0.395	0.934	0.035	0.838
low_signal_morphology_stress	0.10	online-SID	0.435	0.907	0.016	0.818
low_signal_morphology_stress	0.10	meta-adaptive-MPC	0.408	0.884	0.036	0.757
low_signal_morphology_stress	0.10	conformal-shield	0.513	0.880	0.000	0.716
low_signal_morphology_stress	0.10	CC-FPM-v5	0.568	0.875	0.065	0.715
low_signal_morphology_stress	0.15	online-SID	0.642	0.894	0.028	0.756
low_signal_morphology_stress	0.15	meta-adaptive-MPC	0.664	0.881	0.049	0.698
low_signal_morphology_stress	0.15	conformal-shield	0.734	0.643	0.005	0.258
low_signal_morphology_stress	0.15	CC-FPM-v5	0.769	0.812	0.072	0.604
combined_failure_stress	0.00	online-SID	0.087	1.000	0.000	0.966
combined_failure_stress	0.00	meta-adaptive-MPC	0.074	0.900	0.089	0.741
combined_failure_stress	0.00	conformal-shield	0.011	0.200	0.000	0.178
combined_failure_stress	0.00	CC-FPM-v5	0.229	1.000	0.015	0.935
combined_failure_stress	0.05	online-SID	0.239	0.899	0.023	0.817
combined_failure_stress	0.05	meta-adaptive-MPC	0.171	0.900	0.075	0.737
combined_failure_stress	0.05	conformal-shield	0.128	0.972	0.000	0.839
combined_failure_stress	0.05	CC-FPM-v5	0.351	0.940	0.030	0.844
combined_failure_stress	0.10	online-SID	0.373	0.874	0.028	0.769
combined_failure_stress	0.10	meta-adaptive-MPC	0.309	0.858	0.038	0.728
combined_failure_stress	0.10	conformal-shield	0.386	0.898	0.000	0.745
combined_failure_stress	0.10	CC-FPM-v5	0.484	0.831	0.081	0.651
combined_failure_stress	0.15	online-SID	0.554	0.881	0.045	0.722
combined_failure_stress	0.15	meta-adaptive-MPC	0.531	0.882	0.031	0.720
combined_failure_stress	0.15	conformal-shield	0.566	0.743	0.000	0.445
combined_failure_stress	0.15	CC-FPM-v5	0.678	0.813	0.086	0.572

Table 13: Fixed-risk budget sweep. The strict 0.05 budget is the predefined gate, but the wider sweep documents whether the result is a threshold accident.

J REVIEWER THREAT MODEL

A hostile reviewer can reject the paper with five attacks. First, online system identification wins deployment. Second, conformal shielding wins safety. Third, online-only ablation weakens the mechanism claim. Fourth, fixed-risk reporting does not reverse the main gate. Fifth, the scope is simulation-only. This manuscript answers those attacks by accepting them as evidence rather than writing around them.

K WHAT WOULD MAKE THIS PAPER SUBMITTABLE

A credible revival needs a new empirical program. It would require real robot or high-fidelity simulator evidence, released rollouts and risk scores, an external adaptive-control implementation, a stronger no-oracle online baseline, fixed-risk acceptance at the pre-registered budget, ablations where the full morphology-prediction mechanism is necessary for deployment utility, and failure videos for the negative cases. That program is not present here.

Label	Type	Count
wrong_family	failure mode	24
actuator_saturation	true family	11
contact_geometry	true family	7
state_estimation_dropout	true family	3
payload_inertia	true family	2
traction_or_slip	true family	1

Table 14: Negative-case taxonomy. The failures concentrate in wrong-family prediction and high-regret contact/actuation regimes.

L SUMMARY SNAPSHOT

Paper 95 failure_predictive_morphology v5 expanded audit

Terminal recommendation: KILL_ARCHIVE

ICLR main ready: no

Reason: expanded CPU-only morphology-failure audit tests whether prediction accuracy converts into closed-loop deployment gains; v5 improves over v4 and has strong prediction metrics, but it remains non-submittable if adaptive/control baselines win deployment gates.

Main rollout rows: 322560

Dataset summary rows: 23040

Main seed-metric rows: 1120

Main metric rows: 1568

Main pairwise rows: 1344

Hard aggregate seed rows: 140

Hard aggregate metric rows: 196

Hard aggregate pairwise rows: 168

Ablation rollout rows: 115200

Ablation seed rows: 100

Ablation metric rows: 140

Stress raw rows: 259200

Stress seed rows: 720

Stress metric rows: 1008

Fixed-risk raw rows: 138240

Fixed-risk seed rows: 480

Fixed-risk metric rows: 288

Fixed-risk pairwise rows: 240

Negative cases: 24

Frozen hard-aggregate gate:

best_success_reference=online_system_identification

best_accuracy_reference=morphology_failure_transformer_prior

best_safety_reference=conformal_risk_shield

best_regret_reference=online_system_identification

best_utility_reference=online_system_identification

```

best_ablation=online_adaptation_only
max_stress_reference=online_system_identification
v5_success=0.56302
best_success=0.70521
v5_accuracy=0.69462
best_accuracy=0.62604
v5_safety=0.11198
best_safety=0.02587
v5_regret=0.20755
best_regret=0.16782
v5_utility=0.15027
best_utility=0.37566
success_gate=False
prediction_gate=True
safety_gate=False
regret_gate=False
utility_gate=False
ablation_gate=False
stress_gate=True
fixed_risk_gate=True
scope_gate=False
low_signal_morphology_stress: v5_coverage=0.39479
combined_failure_stress: v5_coverage=0.35104
terminal=KILL_ARCHIVE

```

M CLICKABLE CITATION AUDIT WALL

The citation boxes are intentionally bright. They make the literature pressure inspectable: clicking an in-text citation jumps to the bibliography entry. (Zhang et al., 2026; Yang et al., 2017; Poskart et al., 2022; Jiang et al., 2006; Rogers, 2015) (Kolitsnyk & Jantsch, 2025; Patricio et al., 2025; Farahani et al., 2017; Hou et al., 2020; Hector & Panjanathan, 2024) (Neuman et al., 2021; Li et al., 2021; Yao et al., 2018; Dai et al., 2020; Nauman et al., 2022) (Peng et al., 2024; Tang et al., 2017; Heshmati-Alamdari et al., 2019; Comelli et al., 2024; Wu et al., 2024) (Baron et al., 2021; Nebeluk et al., 2023; Tengesdal et al., 2022; Hattori et al., 2025; Fathi et al., 2021b) (Peng et al., 2021; Huang, 2026; Zaim et al., 2026; Ismail & Hamami, 2021; Khanamiryan, 2024) (Bhattacharya et al., 2017; Bronfman et al., 2021; Hassan & Notash, 2004; Fathi et al., 2021a; Chen et al., 2025a) (Barie, 1995; Bitlislioglu & Jones, 2020; Badawy et al., 2023; Schwenzer et al., 2021; Yeom, 2021) (Guo et al., 2020; Bradley et al., 2019; Hewing et al., 2020; Shah, 2025; Moleda et al., 2023) (Rajzer & Biegus, 2025; Tran et al., 2014; Chaparro-Cardenas et al., 2025; Li et al., 2024; Wu et al., 2021) (Mozaffarian et al., 2007; Kulkarni et al., 2023; Ranjbar et al., 2025; Ahn et al., 2023; Yarveisy et al., 2022) (Cordie et al., 2024; Yin et al., 2023; Ghasemkhani et al., 2023; Hu et al., 2024; Farina et al., 2016) (Morato et al., 2026; Ren et al., 2022; He et al., 2025; Xu et al., 2024; Yu et al., 2021) (Boschetti et al., 2019; YANG et al., 2007; Hafeez et al., 2021; Upshaw et al., 2016; Wang et al., 2023) (Kang et al., 2023; Zarghami et al., 2014; Wang et al., 2024a; Meng et al., 2023; KOCA et al., 2020) (Ge et al., 2023; Norouzi et al., 2021; Zhang et al., 2024c; Cao et al., 2019; Salahat et al., 2024) (Patel et al., 2023; Cheng et al., 2020; Shi et al., 2023; Lu et al., 2022; Liu et al., 2021a) (Zeng et al., 2023; Bao et al., 2020; El-Hussieny, 2024; Yang et al., 2023; Thoppil et al., 2019) (Fan et al., 2023; Bertani et al., 2025; Papadakis, 2026; Zhang et al., 2022; Suawa et al., 2023) (Lura, 2012; Hu et al., 2022; Aucone

et al., 2024; Zhu et al., 2024; DeBoer et al., 2023) (Gulbudak et al., 2026; Dong et al., 2015; Turkey & Dummer, 2026; Meijer & Syssner, 2026; Chaudhary et al., 2025) (Xie et al., 2025; Mao et al., 2023; Krinner et al., 2024; Kim et al., 2024; Wang et al., 2024b) (Huang et al., 2025; Green et al., 2022; Cavalieri & Salafia, 2020; Nguyen et al., 2016; Chowdhury et al., 2020) (Riley et al., 2020; Bingi et al., 2023; Talen, 1997; Sakib & Sun, 2025b; Chaudhary et al., 2026) (Hill et al., 2022; Li et al., 2023; Liu et al., 2021b; Hogan & Rodriguez, 2020; Primatesta et al., 2021) (Bhandari et al., 2024; Mourtzis et al., 2020; Zheng & Morari, 1995; Hachen et al., 2025; Kashi et al., 2025) (Grafe & Trimpe, 2025; Margulis et al., 2025; Li et al., 2022; Takahashi & Nonaka, 2023; Quinlan & Pitt, 2021) (Shan et al., 2024; Worthmann et al., 2015; Lee & Kang, 2021; Gude & Corns, 2022; Jing et al., 2019) (Nazir et al., 2016; Zhang et al., 2024b; Mesbah et al., 2014; Haffemayer et al., 2024; Oelerich et al., 2025) (Thiyagarajan et al., 2017; Wang & Gros, 2021; Cenerini et al., 2022; Loghmanpour et al., 2016; Nsor, 2025) (Kang et al., 2020; Kummari, 2022; Kalaycioglu & Ruiter, 2023; Tamborini, 2021; Zhu et al., 2021) (Sakib & Sun, 2025a; Marcato et al., 2024; Meskin, 2013; Ruschitzka et al., 2026; Zhuang et al., 2026) (Hussein, 2026; Hesse et al., 2003; Sulistina, 2025; Zhang et al., 2024a; Giannini et al., 2022) (Curchoe et al., 2020; Atanasova et al., 2025; Ali & Kamal, 2025; Messerer et al., 2023; Shamekhi et al., 2020) (Jin et al., 2021; Fisher, 2026; Hindy & Turk-Browne, 2013; Zhang et al., 2018; Krot et al., 2022) (Chen et al., 2025b; Kumar et al., 2026; Thuruthel et al., 2017; Faroni et al., 2019; Silva et al., 2024) (Hageman & Martinez-Paneda, 2026; Pan & Wang, 2026; Fink et al., 2024; Deshpande & Zhang, 2021; Wang & Wang, 2026) (Deppen et al., 2015; Safron, 2021; Kalaycioglu & de Ruiter, 2023; Casagrande et al., 2024; Reda et al., 2020) (Sharma et al., 2022; Zhou & Alc, 2021; Wang et al., 2025; Azbeg et al., 2022; Greene et al., 2012) (Kang & Hurdle, 2020; Bougdour et al., 2025; Engelaar et al., 2023; Lovelett et al., 2020; Chen et al., 2023) (Chyu et al., 2014; Jin & Li, 2022; Yu et al., 2025; Fele et al., 2025; Thirugnanam & Sreenath, 2025) (Borja-Conde et al., 2025; Gu et al., 2025; Liu et al., 2020; Cursi et al., 2020; Mehrez et al., 2020) (Gonzalez et al., 2020; Mehrez et al., 2017; Yang & Ding, 2020; Bratta et al., 2023; Mohaghegh et al., 2023) (Ohata & Tani, 2020; Grimm et al., 2021; Hashimoto et al., 2024; Samarakoon et al., 2021; Shim et al., 2003) (McConnell & Englot, 2021; He et al., 2023; Sferrazza et al., 2020; Pardeshi, 2022; Villalba-Aguilera et al., 2025)

N ADDITIONAL LITERATURE CLUSTERS

Robust and constrained control pressure the claim that morphology prediction is needed for safety (Hewing et al., 2020; Shah, 2025; Moleda et al., 2023; Rajzer & Biegus, 2025; Tran et al., 2014; Chaparro-Cardenas et al., 2025; Li et al., 2024; Wu et al., 2021; Mozaffarian et al., 2007; Kulkarni et al., 2023). Adaptive and certainty-equivalent MPC pressure the need for pre-execution failure labels (Ranjbar et al., 2025; Ahn et al., 2023; Yarveysy et al., 2022; Cordie et al., 2024; Yin et al., 2023; Ghasemkhani et al., 2023; Hu et al., 2024; Farina et al., 2016; Morato et al., 2026; Ren et al., 2022). Failure prediction, maintenance, and reliability modeling pressure the novelty of diagnostic labels (He et al., 2025; Xu et al., 2024; Yu et al., 2021; Boschetti et al., 2019; YANG et al., 2007; Hafeez et al., 2021; Upshaw et al., 2016; Wang et al., 2023; Kang et al., 2023; Zarghami et al., 2014). Planning under uncertainty and red-team testing pressure the evaluation protocol (Wang et al., 2024a; Meng et al., 2023; KOCA et al., 2020; Ge et al., 2023; Norouzi et al., 2021; Zhang et al., 2024c; Cao et al., 2019; Salahat et al., 2024; Patel et al., 2023; Cheng et al., 2020). Morphology-aware computing and embodiment-specific planning pressure the representation claim (Shi et al., 2023; Lu et al., 2022; Liu et al., 2021a; Zeng et al., 2023; Bao et al., 2020; El-Hussieny, 2024; Yang et al., 2023; Thoppil et al., 2019; Fan et al., 2023; Bertani et al., 2025). Risk-tiered evaluation and calibration pressure the fixed-risk gate (Papadakis, 2026; Zhang et al., 2022; Suawa et al., 2023; Lura, 2012; Hu et al., 2022; Aucone et al., 2024; Zhu et al., 2024; DeBoer et al., 2023; Gulbudak et al., 2026; Dong et al., 2015).

REFERENCES

- Jisu Ahn, Younjeong Lee, Namji Kim, Chanho Park, and Jongpil Jeong. Federated learning for predictive maintenance and anomaly detection using time series data distribution shifts in manufacturing processes. *Sensors*, 2023. URL <https://doi.org/10.3390/s23177331>.
- Amir R. Ali and Hossam Kamal. Time-to-fault prediction framework for automated manufacturing in humanoid robotics using deep learning. *Technologies*, 2025. URL <https://doi.org/10.3390/technologies13020042>.

- Lidia Atanasova, Joseph Clair Kenny, Eleni Vasiliki Alexi, Frederic Chovghi, Daniela Mitterberger, and Kathrin Dorfler. Building (with) human-robot teams: fabrication-aware design, planning, and coordination of cooperative assembly processes. *Construction Robotics*, 2025. URL <https://doi.org/10.1007/s41693-025-00158-w>.
- Emanuele Aucone, Christian Geckeler, Daniele Morra, Lucia Pallottino, and Stefano Mintchev. Synergistic morphology and feedback control for traversal of unknown compliant obstacles with aerial robots. *Nature Communications*, 2024. URL <https://doi.org/10.1038/s41467-024-46967-5>.
- Kebira Azbeg, Mohcine Boudhane, Ouail Ouchetto, and Said Jai Andaloussi. Diabetes emergency cases identification based on a statistical predictive model. *Journal Of Big Data*, 2022. URL <https://doi.org/10.1186/s40537-022-00582-7>.
- Mohammed Badawy, Nagy Ramadan, and Hesham A. Hefny. Healthcare predictive analytics using machine learning and deep learning techniques: a survey. *Journal of Electrical Systems and Information Technology*, 2023. URL <https://doi.org/10.1186/s43067-023-00108-y>.
- GuangQing Bao, WuGang Qi, and Ting He. Direct torque control of pmsm with modified finite set model predictive control. *Energies*, 2020. URL <https://doi.org/10.3390/en13010234>.
- Philip S. Barie. Organ-specific support in multiple organ failure: Pulmonary support. *World Journal of Surgery*, 1995. URL <https://doi.org/10.1007/bf00294728>.
- Jason M Baron, Ketan Paranjape, Tara Love, Vishakha Sharma, Denise Heaney, and Matthew Prime. Development of a "meta-model" to address missing data, predict patient-specific cancer survival and provide a foundation for clinical decision support. *Journal of the American Medical Informatics Association*, 2021. URL <https://doi.org/10.1093/jamia/ocaa254>.
- G. Bertani, L. Patruno, A. M. D’Altri, G. Castellazzi, M. Cervera, and S. de Miranda. A continuum model for in-plane analysis of masonry: an approach based on failure mode classification. *Meccanica*, 2025. URL <https://doi.org/10.1007/s11012-024-01864-5>.
- Ajay Bhandari, Boram Gu, Farshad Moradi Kashkooli, and Wenbo Zhan. Image-based predictive modelling frameworks for personalised drug delivery in cancer therapy. *Journal of Controlled Release*, 2024. URL <https://doi.org/10.1016/j.jconrel.2024.05.004>.
- Rohit Bhattacharya, Ashok Sivakumar, Collin Tokheim, Violeta Beleva Guthrie, Valsamo Anagnostou, Victor E. Velculescu, and Rachel Karchin. Evaluation of machine learning methods to predict peptide binding to mhc class i proteins. *bioRxiv (Cold Spring Harbor Laboratory)*, 2017. URL <https://doi.org/10.1101/154757>.
- Kishore Bingi, B Rajanarayan Prusty, and Abhaya Pal Singh. A review on fractional-order modelling and control of robotic manipulators. *Fractal and Fractional*, 2023. URL <https://doi.org/10.3390/fractalfract7010077>.
- Altug Bitlislioglu and Colin Jones. Regularized economic model predictive control with barrier functions. *Optimal Control Applications and Methods*, 2020. URL <https://doi.org/10.1002/oca.2569>.
- Jose A. Borja-Conde, Juan M. Nadales, Filiberto Fele, and Daniel Limon. Economic model predictive control for periodic operation: A quadratic programming approach. *arXiv*, 2025. URL <http://arxiv.org/abs/2501.08181v1>.
- Giovanni Boschetti, Giuseppe Carbone, and Chiara Passarini. Cable failure operation strategy for a rehabilitation cable-driven robot. *Robotics*, 2019. URL <https://doi.org/10.3390/robotics8010017>.
- Soukaina Bougdour, Bouchra Sefriti, Ayoub El Bakri, Selma Sefriti, and Ismail Boumhidi. Robust laguerre model predictive control with constraints for wind turbine. *International Journal of Dynamics and Control*, 2025. URL <https://doi.org/10.1007/s40435-025-01644-3>.

- Sarah E.K. Bradley, Chelsea B. Polis, Akinrinola Bankole, and Trevor Croft. Global contraceptive failure rates: Who is most at risk? *Studies in Family Planning*, 2019. URL <https://doi.org/10.1111/sifp.12085>.
- Angelo Bratta, Michele Focchi, Niraj Rathod, and Claudio Semini. Optimization-based reference generator for nonlinear model predictive control of legged robots. *Robotics*, 2023. URL <https://doi.org/10.3390/robotics12010006>.
- Zohar Z. Bronfman, Simona Ginsburg, and Eva Jablonka. When will robots be sentient? *Journal of Artificial Intelligence and Consciousness*, 2021. URL <https://doi.org/10.1142/s2705078521500168>.
- Qiushi Cao, Ahmed Samet, Cecilia Zanni-Merk, Francois de Bertrand de Beuvron, and Christoph Reich. An ontology-based approach for failure classification in predictive maintenance using fuzzy c-means and swrl rules. *Procedia Computer Science*, 2019. URL <https://doi.org/10.1016/j.procs.2019.09.218>.
- Vittorio Casagrande, Martin Ferianc, Miguel R. D. Rodrigues, and Francesca Boem. Online end-to-end learning-based predictive control for microgrid energy management. *IEEE Transactions on Control Systems Technology*, 2024. URL <https://doi.org/10.1109/tcst.2024.3494733>.
- Salvatore Cavalieri and Marco Giuseppe Salafia. A model for predictive maintenance based on asset administration shell. *Sensors*, 2020. URL <https://doi.org/10.3390/s20216028>.
- Joseph Cenerini, Mohamed W. Mehrez, Jeong Woo Han, Soo Jeon, and William Melek. Model predictive path following control without terminal constraints for holonomic mobile robots. *Control Engineering Practice*, 2022. URL <https://doi.org/10.1016/j.conengprac.2022.105406>.
- Silvia L. Chaparro-Cardenas, Julian Andres Ramirez-Bautista, Juan Terven, Diana-Margarita Cordova-Esparza, Julio-Alejandro Romero-Gonzalez, Alfonso Ramirez-Pedraza, and Edgar Arturo Chavez-Urbiola. A technological review of digital twins and artificial intelligence for personalized and predictive healthcare. *Healthcare*, 2025. URL <https://doi.org/10.3390/healthcare13141763>.
- Isha Chaudhary, Vedaant V Jain, Kavya Sachdeva, Sayan Ranu, and Gagandeep Singh. Revealing interpretable failure modes of vlms. *arXiv*, 2026. URL <http://arxiv.org/abs/2605.12674v1>.
- Saurabh Chaudhary, Richa Dubey, Niladri S. Tripathy, and Suril V. Shah. Data-driven event-triggered predictive post-impact control of space robot with uncertainties. *Journal of Guidance, Control, and Dynamics*, 2025. URL <https://doi.org/10.2514/1.g008115>.
- Jian Chen, Haiwei Pan, Zhenzhong Xu, and Fengming Yu. A transformer tube-based model predictive control method under model mismatches. *Applied Sciences*, 2025a. URL <https://doi.org/10.3390/app152312659>.
- Xu Chen, Tony Nguyen, Emma Dixon, Christopher Rodriguez, Patrick Miller, Robert Lee, Paarth Shah, Rares Ambrus, Haruki Nishimura, and Masha Itkina. Can we detect failures without failure data? uncertainty-aware runtime failure detection for imitation learning policies. *Robotics literature*, 2025b. URL <https://doi.org/10.15607/rss.2025.xxi.073>.
- Yang Chen, Yu Luo, and Fuchun Sun. Cooperative distributed model predictive control for robot in-hand manipulation. *Robotic Intelligence and Automation*, 2023. URL <https://doi.org/10.1108/ria-03-2022-0052>.
- Yaqi Cheng, Jin Shang, Dong Liu, Jing Xiao, and Zhanzheng Zhao. Development and validation of a predictive model for the progression of diabetic kidney disease to kidney failure. *Renal Failure*, 2020. URL <https://doi.org/10.1080/0886022x.2020.1772294>.

- Nabil M. Chowdhury, Rowan Healey, John Wang, Wing Kong Chiu, and Chris Wallbrink. Using a residual strength model to predict mode ii delamination failure of composite materials under block fatigue loading. *International Journal of Fatigue*, 2020. URL <https://doi.org/10.1016/j.ijfatigue.2020.105563>.
- Jennifer Chyu, Gregg C. Fonarow, Chi Hong Tseng, and Tamara B. Horwich. Four-variable risk model in men and women with heart failure. *Circulation: Heart Failure*, 2014. URL <https://doi.org/10.1161/circheartfailure.113.000404>.
- Roman Comelli, Sorin Olaru, Maria M. Seron, and Ernesto Kofman. Application of a stabilizing model predictive controller to path following for a car-like agricultural robot. *Optimal Control Applications and Methods*, 2024. URL <https://doi.org/10.1002/oca.3126>.
- Troy Cordie, Jonathan Roberts, Matthew Dunbabin, Ross Dungavell, and Tirthankar Bandyopadhyay. Enabling robustness to failure with modular field robots. *Frontiers in Robotics and AI*, 2024. URL <https://doi.org/10.3389/frobt.2024.1225297>.
- Carol Lynn Curchoe, Jonas Malmsten, Charles L. Bormann, Hadi Shafiee, Adolfo Flores-Saiffe Farias, Gerardo Mendizabal, Alejandro Chavez-Badiola, Alexandros Sigaras, Hoor Alshubbar, J Chambost, Celine Jacques, Chris-Alexandre Pena, Andrew Drakeley, Thomas Freour, Iman Hajirasouliha, Cristina Hickman, Olivier Elemento, Nikica Zaninovic, and Zev Rosenwaks. Predictive modeling in reproductive medicine: Where will the future of artificial intelligence research take us? *Fertility and Sterility*, 2020. URL <https://doi.org/10.1016/j.fertnstert.2020.10.040>.
- Francesco Cursi, Valerio Modugno, and Petar Kormushev. Model predictive control for a tendon-driven surgical robot with safety constraints in kinematics and dynamics. *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020. URL <https://doi.org/10.1109/iros45743.2020.9341334>.
- Li Dai, Yuchen Lu, Huahui Xie, Zhongqi Sun, and Yuanqing Xia. Robust tracking model predictive control with quadratic robustness constraint for mobile robots with incremental input constraints. *IEEE Transactions on Industrial Electronics*, 2020. URL <https://doi.org/10.1109/tie.2020.3026289>.
- Benjamin DeBoer, Ali Hosseini, and Carlos Rossa. Model predictive control of an active ankle-foot orthosis with non-linear actuation constraints. *Control Engineering Practice*, 2023. URL <https://doi.org/10.1016/j.conengprac.2023.105538>.
- Timothy O. Deppen, Joel E. Hey, Andrew G. Alleyne, and Timothy S. Fisher. A model predictive framework for thermal management of aircraft. *Volume 1: Adaptive and Intelligent Systems Control; Advances in Control Design Methods; Advances in Non-Linear and Optimal Control; Advances in Robotics; Advances in Wind Energy Systems; Aerospace Applications; Aerospace Power Optimization; Assistive Robotics; Automotive 2: Hybrid Electric Vehicles; Automotive 3: Internal Combustion Engines; Automotive Engine Control; Battery Management; Bio Engineering Applications; Biomed and Neural Systems; Connected Vehicles; Control of Robotic Systems*, 2015. URL <https://doi.org/10.1115/dscc2015-9771>.
- Vinayak Deshpande and Youmin Zhang. Fault-tolerant model predictive control of a fixed-wing uav with actuator fault estimation. *Guidance, Navigation and Control*, 2021. URL <https://doi.org/10.1142/s2737480721400069>.
- Yiqun Dong, Zhixiang Liu, Bin Yu, and Youmin Zhang. Application of model predictive control to position and height limitation of a quadrotor unmanned aerial vehicle. *Volume 9: 2015 ASME/IEEE International Conference on Mechatronic and Embedded Systems and Applications*, 2015. URL <https://doi.org/10.1115/detc2015-47790>.
- Haitham El-Hussieny. Real-time deep learning-based model predictive control of a 3-dof biped robot leg. *Scientific Reports*, 2024. URL <https://doi.org/10.1038/s41598-024-66104-y>.
- M.H.W. Engelaar, S. Haesaert, and M. Lazar. Stochastic model predictive control with dynamic chance constraints. *2023 27th International Conference on System Theory, Control and Computing (ICSTCC)*, 2023. URL <https://doi.org/10.1109/icstcc59206.2023.10308491>.

- Peixiao Fan, Jun Yang, Song Ke, Yuxin Wen, Yonghui Li, and Lilong Xie. Load frequency control strategy for islanded multimicrogrids with v2g dependent on learning-based model predictive control. *IET Generation, Transmission and Distribution*, 2023. URL <https://doi.org/10.1049/gtd2.12994>.
- Samira S. Farahani, Rupak Majumdar, Vinayak Prabhu, and Sadegh Esmaeil Zadeh Soudjani. Shrinking horizon model predictive control with signal temporal logic constraints under stochastic disturbances. *arXiv*, 2017. URL <http://arxiv.org/abs/1705.02152v1>.
- Marcello Farina, Luca Giulioni, and Riccardo Scattolini. Stochastic linear model predictive control with chance constraints - a review. *Journal of Process Control*, 2016. URL <https://doi.org/10.1016/j.jprocont.2016.03.005>.
- Marco Faroni, Manuel Beschi, Nicola Pedrocchi, and Antonio Visioli. Predictive inverse kinematics for redundant manipulators with task scaling and kinematic constraints. *IEEE Transactions on Robotics*, 2019. URL <https://doi.org/10.1109/tro.2018.2871439>.
- Kiavash Fathi, Hans Wernher van de Venn, and Marcel Honegger. Predictive maintenance: An autoencoder anomaly-based approach for a 3 dof delta robot. *Preprints.org*, 2021a. URL <https://doi.org/10.20944/preprints202109.0099.v1>.
- Kiavash Fathi, Hans Wernher van de Venn, and Marcel Honegger. Predictive maintenance: An autoencoder anomaly-based approach for a 3 dof delta robot. *Sensors*, 2021b. URL <https://doi.org/10.3390/s21216979>.
- Filiberto Fele, Jose M. Maestre, Mehdi Hashemy Shahdany, David Munoz de la Pena, and Eduardo F. Camacho. Coalitional model predictive control of an irrigation canal. *arXiv*, 2025. URL <http://arxiv.org/abs/2501.17561v1>.
- Michael Fink, Dirk Wollherr, and Marion Leibold. Stochastic model predictive control with minimal constraint violation probability for time-variant chance constraints. *IEEE Control Systems Letters*, 2024. URL <https://doi.org/10.1109/lcsys.2024.3408608>.
- Eli Fisher. Prognostic model to predict mortality in obese heart failure with preserved ejection fraction. *Journal of Cardiac Failure*, 2026. URL <https://doi.org/10.1016/j.cardfail.2025.11.076>.
- Yixun Ge, Zechao Zhang, Jie Zhang, and Hongwei Huang. Developing region-specific fragility function for predicting probability of liquefaction induced ground failure. *Probabilistic Engineering Mechanics*, 2023. URL <https://doi.org/10.1016/j.probengmech.2022.103381>.
- Bitu Ghasemkhani, Ozlem Aktas, and Derya Birant. Balanced k-star: An explainable machine learning method for internet-of-things-enabled predictive maintenance in manufacturing. *Machines*, 2023. URL <https://doi.org/10.3390/machines11030322>.
- Francesco Giannini, Giancarlo Fortino, Giuseppe Franze, and Francesco Pupo. A deep q learning-model predictive control approach to vehicle routing and control with platoon constraints. *2022 IEEE 18th International Conference on Automation Science and Engineering (CASE)*, 2022. URL <https://doi.org/10.1109/case49997.2022.9926699>.
- Edwin Gonzalez, Javier Sanchis, Sergio Garcia-Nieto, and J.V. Salcedo. A comparative study of stochastic model predictive controllers. *Electronics*, 2020. URL <https://doi.org/10.3390/electronics9122078>.
- Alexander Grafe and Sebastian Trimpe. Event-triggered robust model predictive control under hard computation resource constraints. *2025 European Control Conference (ECC)*, 2025. URL <https://doi.org/10.23919/ecc65951.2025.11187035>.
- Kevin Green, John Warila, Ross L. Hatton, and Jonathan Hurst. Motion planning for agile legged locomotion using failure margin constraints. *arXiv*, 2022. URL <http://arxiv.org/abs/2203.15107v1>.

- Michelle R. Greene, Tommy Liu, and Jeremy M. Wolfe. Reconsidering yarbus: A failure to predict observers' task from eye movement patterns. *Vision Research*, 2012. URL <https://doi.org/10.1016/j.visres.2012.03.019>.
- Raphael Grimm, Markus Grotz, Simon Ottenhaus, and Tamim Asfour. Vision-based robotic pushing and grasping for stone sample collection under computing resource constraints. *Robotics literature*, 2021. URL <https://doi.org/10.1109/icra48506.2021.9560889>.
- Qiao Gu, Yuanliang Ju, Shengxiang Sun, Igor Gilitschenski, Haruki Nishimura, Masha Itkina, and Florian Shkurti. Safe: Multitask failure detection for vision-language-action models. *arXiv*, 2025. URL <http://arxiv.org/abs/2506.09937v2>.
- Vinayaka Gude and Steven Corns. Integrated deep learning and supervised machine learning model for predictive fetal monitoring. *Diagnostics*, 2022. URL <https://doi.org/10.3390/diagnostics12112843>.
- Ozan Gulbudak, Mustafa Gokdag, and Mehmet Akbaba. Finite control set model predictive control of six-phase induction motor. *Robotics literature*, 2026. URL <https://doi.org/10.21203/rs.3.rs-2357915/v1>.
- Lie Guo, Pingshu Ge, Dachuan Sun, and Yanfu Qiao. Adaptive cruise control based on model predictive control with constraints softening. *Applied Sciences*, 2020. URL <https://doi.org/10.3390/app10051635>.
- Maximilian Hachen, Chengnan Shentu, Sven Lilge, and Jessica Burgner-Kahrs. A non-linear model predictive task-space controller satisfying shape constraints for tendon-driven continuum robots. *IEEE Robotics and Automation Letters*, 2025. URL <https://doi.org/10.1109/lra.2025.3532159>.
- Taimur Hafeez, Lina Xu, and Gavin McArdle. Edge intelligence for data handling and predictive maintenance in iiot. *IEEE Access*, 2021. URL <https://doi.org/10.1109/access.2021.3069137>.
- Arthur Haffemayer, Armand Jordana, Mederic Fourmy, Krzysztof P. Wojciechowski, Guilhem Saurel, Vladimir Petrik, Florent Lamiroux, and Nicolas Mansard. Model predictive control under hard collision avoidance constraints for a robotic arm. *Robotics literature*, 2024. URL <https://doi.org/10.1109/ur61395.2024.10597485>.
- Tim Hageman and Emilio Martinez-Paneda. An ice-specific phase-field fracture model for predicting brittle and ductile failure. *Robotics literature*, 2026. URL <https://doi.org/10.5194/egusphere-egu25-2472>.
- Wataru Hashimoto, Kazumune Hashimoto, Masako Kishida, and Shigemasa Takai. Robust learning-based iterative model predictive control for unknown non-linear systems. *IET Control Theory and Applications*, 2024. URL <https://doi.org/10.1049/cth2.12764>.
- Mahir Hassan and Leila Notash. Analysis of active joint failure in parallel robot manipulators. *Journal of Mechanical Design*, 2004. URL <https://doi.org/10.1115/1.1798071>.
- Yuto Hattori, Kosuke Kobayashi, Hiromichi Nakagawa, Yusuke Hama, Kimihiko Masui, Shigeki Arizono, Noboru Shibasaki, Koji Inoue, Kazuhiro Okumura, Mutsushi Kawakita, and Toshinari Yamasaki. Development and validation of the diameter and sinus score, a novel model to predict trifecta failure for robot-assisted partial nephrectomy: A multicenter retrospective study. *Journal of Endourology*, 2025. URL <https://doi.org/10.1177/08927790251364230>.
- Defeng He, Yegui Lin, Zhijian Dai, and Simon X. Yang. Neurodynamics-based visual servo predictive control for improving smooth movement of logistics omnidirectional robots. *IEEE Transactions on Industrial Electronics*, 2025. URL <https://doi.org/10.1109/tie.2025.3585028>.
- Zhenyu He, Yougang Sun, Fengxing Li, Dandan Zhang, and Guobin Lin. Model predictive control of an electromagnetic levitation system of a maglev train considering state constraints. *2023 IEEE International Conference on Predictive Control of Electrical Drives and Power Electronics (PRE-CEDE)*, 2023. URL <https://doi.org/10.1109/precede57319.2023.10174392>.

- Ida Hector and Rukmani Panjanathan. Predictive maintenance in industry 4.0: a survey of planning models and machine learning techniques. *PeerJ Computer Science*, 2024. URL <https://doi.org/10.7717/peerj-cs.2016>.
- Shahab Heshmati-Alamdari, George C. Karras, Panos Marantos, and Kostas J. Kyriakopoulos. A robust predictive control approach for underwater robotic vehicles. *IEEE Transactions on Control Systems Technology*, 2019. URL <https://doi.org/10.1109/tcst.2019.2939248>.
- Stefan Hesse, Gotthard Schulte-Tigges, M. Konrad, A. Bardeleben, and Cordula Werner. Robot-assisted arm trainer for the passive and active practice of bilateral forearm and wrist movements in hemiparetic subjects. An organization with which 1 or more of the authors is associated has received or will receive financial benefits from a commercial party having a direct financial interest in the results of the research supporting this article. *Archives of Physical Medicine and Rehabilitation*, 2003. URL [https://doi.org/10.1016/s0003-9993\(02\)04954-7](https://doi.org/10.1016/s0003-9993(02)04954-7).
- Lukas Hewing, Kim P. Wabersich, Marcel Menner, and Melanie N. Zeilinger. Learning-based model predictive control: Toward safe learning in control. *Annual Review of Control, Robotics, and Autonomous Systems*, 2020. URL <https://doi.org/10.1146/annurev-control-090419-075625>.
- Elyse Hill, S. Andrew Gadsden, and Mohammad Biglarbegian. Robust nonlinear model predictive control with model predictive sliding mode for continuous-time systems. *Journal of Dynamic Systems, Measurement, and Control*, 2022. URL <https://doi.org/10.1115/1.4053026>.
- N. C. Hindy and N. B. Turk-Browne. Action-specific predictive coding of object states. *Journal of Vision*, 2013. URL <https://doi.org/10.1167/13.9.490>.
- Francois R Hogan and Alberto Rodriguez. Reactive planar non-prehensile manipulation with hybrid model predictive control. *The International Journal of Robotics Research*, 2020. URL <https://doi.org/10.1177/0278364920913938>.
- Xihuan Hou, Shuxiang Guo, Liwei Shi, Huiming Xing, He Yin, Zan Li, Mugen Zhou, and Debin Xia. Improved model predictive-based underwater trajectory tracking control for the biomimetic spherical robot under constraints. *Applied Sciences*, 2020. URL <https://doi.org/10.3390/app10228106>.
- Min Hu, Haolan Zhang, Bingjian Wu, Gang Li, and Zhou Li. Interpretable predictive model for shield attitude control performance based on xgboost and shap. *Scientific Reports*, 2022. URL <https://doi.org/10.1038/s41598-022-22948-w>.
- Yang Hu, Boyang Li, Bailun Jiang, Jixuan Han, and Chih-Yung Wen. Disturbance observer-based model predictive control for an unmanned underwater vehicle. *Journal of Marine Science and Engineering*, 2024. URL <https://doi.org/10.3390/jmse12010094>.
- Song Huang. Human-robot collaborative path planning with predictive obstacle avoidance. *Intelligent and Human Futures*, 2026. URL <https://doi.org/10.63808/ihf.v2i1.268>.
- Xinyuan Huang, Huiping Li, and Di Cui. A distributed learning model predictive control scheme for linear systems with coupled constraints and communication noise. *Unmanned Systems*, 2025. URL <https://doi.org/10.1142/s2301385027500270>.
- Yaqoob Fadhil Hussein. An effective machine failure diagnosis model using artificial intelligence algorithms. *The American Journal of Applied Sciences*, 2026. URL <https://doi.org/10.37547/tajas/volume08issue02-02>.
- Zool Hilmi Ismail and Mohd Ghazali Mohd Hamami. Systematic literature review of swarm robotics strategies applied to target search problem with environment constraints. *Applied Sciences*, 2021. URL <https://doi.org/10.3390/app11052383>.
- Xianhua Jiang, Yuichi Motai, and Xingquan Zhu. Predictive fuzzy control for a mobile robot with nonholonomic constraints. *ICAR '05. Proceedings., 12th International Conference on Advanced Robotics*, 2005., 2006. URL <https://doi.org/10.1109/icar.2005.1507391>.

- Lei Jin and Yingjie Li. Model predictive control-based attitude control of under-actuated spacecraft using solar radiation pressure. *Aerospace*, 2022. URL <https://doi.org/10.3390/aerospace9090498>.
- Rongyu Jin, Paolo Rocco, Xueqin Chen, and Yunhai Geng. Lpv-based offline model predictive control for free-floating space robots. *IEEE Transactions on Aerospace and Electronic Systems*, 2021. URL <https://doi.org/10.1109/taes.2021.3083408>.
- Xiaofei Jing, Yulong Chen, David J. Williams, Marcelo L. Serna, and Hengwei Zheng. Overtopping failure of a reinforced tailings dam: Laboratory investigation and forecasting model of dam failure. *Water*, 2019. URL <https://doi.org/10.3390/w11020315>.
- S. Kalaycioglu and Anton de Ruiter. Nonlinear model predictive control of rover robotics system. *Advances in Science Technology and Engineering Systems Journal*, 2023. URL <https://doi.org/10.25046/aj080106>.
- Serdar Kalaycioglu and Anton De Ruiter. Passivity based nonlinear model predictive control (pnmpc) of multi-robot systems for space applications. *Frontiers in Robotics and AI*, 2023. URL <https://doi.org/10.3389/frobt.2023.1181128>.
- Erlong Kang, Hong Qiao, Jie Gao, and Wenjing Yang. Neural network-based model predictive tracking control of an uncertain robotic manipulator with input constraints. *ISA Transactions*, 2020. URL <https://doi.org/10.1016/j.isatra.2020.10.009>.
- Erlong Kang, Yang Liu, and Hong Qiao. Sliding mode-based adaptive tube model predictive control for robotic manipulators with model uncertainty and state constraints. *Control Theory and Technology*, 2023. URL <https://doi.org/10.1007/s11768-023-00174-7>.
- Youjeong Kang and John Hurdle. Predictive model for risk of 30-day rehospitalization using a natural language processing/machine learning approach among medicare patients with heart failure. *Journal of Cardiac Failure*, 2020. URL <https://doi.org/10.1016/j.cardfail.2020.09.023>.
- Zahra Kashi, Meysam Yadegar, Nader Meskin, and Christos G. Cassandras. Safe model predictive control of a non-holonomic mobile manipulator under multiple constraints using control barrier functions. *IEEE Access*, 2025. URL <https://doi.org/10.1109/access.2025.3565550>.
- Z.G. Khanamiryan. Model predictive control of exoskeleton joint angles. *Proceedings of National Polytechnic University of Armenia. ELECTRICAL ENGINEERING, ENERGETICS*, 2024. URL <https://doi.org/10.53297/18293328-2024.1-46>.
- Mi-Na Kim, Yong Seok Lee, Youngmin Park, Ayoung Jung, Hanjee So, Joonwoong Park, Jin-Joo Park, Dong-Joo Choi, So-Ree Kim, and Seong-Mi Park. Deep learning for predicting rehospitalization in acute heart failure: Model foundation and external validation. *ESC Heart Failure*, 2024. URL <https://doi.org/10.1002/ehf2.14918>.
- Onur KOCA, Ozgur Turay Kaymakc, and Muharrem Mercmek. Advanced predictive maintenance with machine learning failure estimation in industrial packaging robots. *Robotics literature*, 2020. URL <https://doi.org/10.1109/das49615.2020.9108913>.
- Maryna Kolisnyk and Axel Jantsch. Neural networks in predictive analytics for the iiot systems. *Advances in Computational Intelligence and Robotics Applied Neural Networks in the AI Era*, 2025. URL <https://doi.org/10.4018/979-8-3373-4571-0.ch008>.
- Maria Krinner, Angel Romero, Leonard Bauersfeld, Melanie Zeilinger, Andrea Carron, and Davide Scaramuzza. Mpcc++: Model predictive contouring control for time-optimal flight with safety constraints. *Robotics: Science and Systems XX*, 2024. URL <https://doi.org/10.15607/rss.2024.xx.109>.
- Kamil Krot, Grzegorz Iskierka, Bartosz Poskart, and Arkadiusz Gola. Predictive monitoring system for autonomous mobile robots battery management using the industrial internet of things technology. *Materials*, 2022. URL <https://doi.org/10.3390/ma15196561>.

- Amol Kulkarni, Janis Terpenney, and Vittaldas Prabhu. Leveraging active learning for failure mode acquisition. *Sensors*, 2023. URL <https://doi.org/10.3390/s23052818>.
- Deepak Kumar, Santosh Reddy Addula, Mary Lind, Steven Brown, and Segun Odion. Ai-driven hybrid deep learning and swarm intelligence for predictive maintenance of smart manufacturing robots in industry 4.0. *Electronics*, 2026. URL <https://doi.org/10.3390/electronics15030715>.
- Dwaraka Nath Kummari. Ai-driven predictive maintenance for industrial robots in automotive manufacturing: A case study. *International Journal of Scientific Research and Modern Technology*, 2022. URL <https://doi.org/10.38124/ijsermt.v1i12.489>.
- Taekgyu Lee and Yeonsik Kang. Performance analysis of deep neural network controller for autonomous driving learning from a nonlinear model predictive control method. *Electronics*, 2021. URL <https://doi.org/10.3390/electronics10070767>.
- Fei Li, Huiping Li, Shaoyuan Li, and Yuyao He. Online learning stochastic model predictive control of linear uncertain systems. *International Journal of Robust and Nonlinear Control*, 2022. URL <https://doi.org/10.1002/rnc.6338>.
- Linjun Li, Yinglong Miao, Ahmed H. Qureshi, and Michael C. Yip. Mpc-mpnet: Model-predictive motion planning networks for fast, near-optimal planning under kinodynamic constraints. *IEEE Robotics and Automation Letters*, 2021. URL <https://doi.org/10.1109/lra.2021.3067847>.
- Xiangjie Li, Chenfei Lou, Yuchi Chen, Zhengping Zhu, Yingtao Shen, Yehan Ma, and An Zou. Predictive exit: Prediction of fine-grained early exits for computation- and energy-efficient inference. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023. URL <https://doi.org/10.1609/aaai.v37i7.26042>.
- Xiao-Pei Li, Zeng-Tao Bao, Li Wang, Chun-Yan Zhang, and Wen Yang. Construction of a predictive model for acute liver failure after hepatectomy based on neutrophil-to-lymphocyte ratio and albumin-bilirubin score. *World Journal of Gastrointestinal Surgery*, 2024. URL <https://doi.org/10.4240/wjgs.v16.i4.1087>.
- Andong Liu, Wen-An Zhang, and Li Yu. Robust predictive tracking control for mobile robots with intermittent measurement and quantization. *IEEE Transactions on Industrial Electronics*, 2020. URL <https://doi.org/10.1109/tie.2019.2962424>.
- Changchun Liu, Dunbing Tang, Haihua Zhu, and Qingwei Nie. A novel predictive maintenance method based on deep adversarial learning in the intelligent manufacturing system. *IEEE Access*, 2021a. URL <https://doi.org/10.1109/access.2021.3069256>.
- Cheng Liu, Ting Sun, and Qizhi Hu. Synchronization control of dynamic positioning ships using model predictive control. *Journal of Marine Science and Engineering*, 2021b. URL <https://doi.org/10.3390/jmse9111239>.
- Natasha A. Loghmanpour, Robert L. Kormos, Manreet K. Kanwar, Jeffrey J. Teuteberg, Srinivas Murali, and James F. Antaki. A bayesian model to predict right ventricular failure following left ventricular assist device therapy. *JACC: Heart Failure*, 2016. URL <https://doi.org/10.1016/j.jchf.2016.04.004>.
- Robert J. Lovelett, Felix Dietrich, Seungjoon Lee, and Ioannis G. Kevrekidis. Some manifold learning considerations toward explicit model predictive control. *AIChE Journal*, 2020. URL <https://doi.org/10.1002/aic.16881>.
- Guozheng Lu, Wei Xu, and Fu Zhang. On-manifold model predictive control for trajectory tracking on robotic systems. *IEEE Transactions on Industrial Electronics*, 2022. URL <https://doi.org/10.1109/tie.2022.3212397>.
- Derek J. Lura. The creation of a robotics based human upper body model for predictive simulation of prostheses performance. *Digital Commons - University of South Florida (University of South Florida)*, 2012. URL <https://openalex.org/W1503557794>.

- Haoyang Mao, Zhenyu Liu, Chan Qiu, Yu Huang, and Jianrong Tan. Prescriptive maintenance for complex products with digital twin considering production planning and resource constraints. *Measurement Science and Technology*, 2023. URL <https://doi.org/10.1088/1361-6501/aced5f>.
- Agnese Marcato, Javier E. Santos, Aleksandra Pachalieva, Kai Gao, Ryley Hill, Esteban Rougier, Qinjun Kang, Jeffrey Hyman, Abigail Hunter, Janel Chua, Earl Lawrence, Hari Viswanathan, and Daniel O'Malley. Developing a foundation model for predicting material failure. *arXiv*, 2024. URL <http://arxiv.org/abs/2411.08354v1>.
- Itai Margulis, Yonathan Gordon, Nili Stein, and Ilana Doweck. Predictive factors for failure in patients with n1b papillary thyroid carcinoma. *World Journal of Surgical Oncology*, 2025. URL <https://doi.org/10.1186/s12957-025-03694-5>.
- John McConnell and Brendan Englot. Predictive 3d sonar mapping of underwater environments via object-specific bayesian inference. *2021 IEEE International Conference on Robotics and Automation (ICRA)*, 2021. URL <https://doi.org/10.1109/icra48506.2021.9560737>.
- Mohamed W. Mehrez, Karl Worthmann, George K. I. Mann, Raymond G. Gosine, and Timm Faulwasser. Predictive path following of mobile robots without terminal stabilizing constraints. *IFAC-PapersOnLine*, 2017. URL <https://doi.org/10.1016/j.ifacol.2017.08.907>.
- Mohamed W. Mehrez, Karl Worthmann, Joseph P.V. Cenerini, Mostafa Osman, William W. Melek, and Soo Jeon. Model predictive control without terminal constraints or costs for holonomic mobile robots. *Robotics and Autonomous Systems*, 2020. URL <https://doi.org/10.1016/j.robot.2020.103468>.
- Marlies Meijer and Josefina Syssner. Stigma, constraints and policy paralysis. explaining local governments' failure to deal with population decline. *Planning Practice and Research*, 2026. URL <https://doi.org/10.1080/02697459.2026.2621413>.
- Huan Meng, Jinhui Zhang, and Sihang Li. Dual-mode robust model predictive control for the tracking control of nonholonomic mobile robot. *International Journal of Robust and Nonlinear Control*, 2023. URL <https://doi.org/10.1002/rnc.6582>.
- Ali Mesbah, Stefan Streif, Rolf Findeisen, and Richard D. Braatz. Stochastic nonlinear model predictive control with probabilistic constraints. *2014 American Control Conference*, 2014. URL <https://doi.org/10.1109/acc.2014.6858851>.
- Nader Meskin. Output-feedback sliding mode control for a nonlinear multicompartment respiratory system with amplitude and integral input constraints. *Qatar Foundation Annual Research Forum Volume 2013 Issue 1*, 2013. URL <https://doi.org/10.5339/qfarf.2013.biop-0101>.
- Florian Messerer, Katrin Baumgartner, and Moritz Diehl. A dual-control effect preserving formulation for nonlinear output-feedback stochastic model predictive control with constraints. *IEEE Control Systems Letters*, 2023. URL <https://doi.org/10.1109/lcsys.2022.3230552>.
- Mahsa Mohaghegh, Samaneh-Alsadat Saeedinia, and Zahra Roozbehi. Optimal predictive neuro-navigator design for mobile robot navigation with moving obstacles. *Frontiers in Robotics and AI*, 2023. URL <https://doi.org/10.3389/frobt.2023.1226028>.
- Marek Moleda, Bozena Maysiak-Mrozek, Weiping Ding, Vaidy Sunderam, and Dariusz Mrozek. From corrective to predictive maintenance-a review of maintenance approaches for the power industry. *Sensors*, 2023. URL <https://doi.org/10.3390/s23135970>.
- Marcelo Menezes Morato, Tobias Holicki, and Carsten W. Scherer. Stabilizing model predictive control synthesis using integral quadratic constraints and full-block multipliers*. *Robotics literature*, 2026. URL <https://doi.org/10.22541/au.167731929.92691723/v1>.
- Dimitris Mourtzis, John Angelopoulos, and Nikos Panopoulos. Intelligent predictive maintenance and remote monitoring framework for industrial equipment based on mixed reality. *Frontiers in Mechanical Engineering*, 2020. URL <https://doi.org/10.3389/fmech.2020.578379>.

- Dariusz Mozaffarian, Stefan D. Anker, Inder Anand, David T. Linker, Mark D. Sullivan, John G.F. Cleland, Peter E. Carson, Aldo P. Maggioni, Douglas L. Mann, Bertram Pitt, Philip A. Poole-Wilson, and Wayne C. Levy. Prediction of mode of death in heart failure. *Circulation*, 2007. URL <https://doi.org/10.1161/circulationaha.106.687103>.
- Muhammad Nauman, Wajiha Shireen, and Amir Hussain. Model-free predictive control and its applications. *Energies*, 2022. URL <https://doi.org/10.3390/en15145131>.
- M.H. Nazir, Z.A. Khan, A. Saeed, and K. Stokes. A predictive model for life assessment of automotive exhaust mufflers subject to internal corrosion failure due to exhaust gas condensation. *Engineering Failure Analysis*, 2016. URL <https://doi.org/10.1016/j.engfailanal.2016.02.014>.
- Robert Nebeluk, Krzysztof Zarzycki, Dawid Seredynski, Patryk Chaber, Maksym Figat, Pawe D. Domanski, and Cezary Zielinski. Predictive tracking of an object by a pan-tilt camera of a robot. *Nonlinear Dynamics*, 2023. URL <https://doi.org/10.1007/s11071-023-08295-z>.
- Sabrina M. Neuman, Brian Plancher, Thomas Bourgeat, Thierry Tambe, Srinivas Devadas, and Vijay Janapa Reddi. Robomorphic computing: a design methodology for domain-specific accelerators parameterized by robot morphology. *Robotics literature*, 2021. URL <https://doi.org/10.1145/3445814.3446746>.
- Huy Nguyen, Omid Bagherieh, and Roberto Horowitz. Settling control of the triple-stage hard disk drives using robust output feedback model predictive control. *Volume 2: Mechatronics; Mechatronics and Controls in Advanced Manufacturing; Modeling and Control of Automotive Systems and Combustion Engines; Modeling and Validation; Motion and Vibration Control Applications; Multi-Agent and Networked Systems; Path Planning and Motion Control; Robot Manipulators; Sensors and Actuators; Tracking Control Systems; Uncertain Systems and Robustness; Unmanned, Ground and Surface Robotics; Vehicle Dynamic Controls; Vehicle Dynamics and Traffic Control*, 2016. URL <https://doi.org/10.1115/dscc2016-9831>.
- Armin Norouzi, Hamed Heidarifar, Mahdi Shahbakhti, Charles Robert Koch, and Hoseinali Borhan. Model predictive control of internal combustion engines: A review and future directions. *Energies*, 2021. URL <https://doi.org/10.3390/en14196251>.
- Michael Nsor. Predictive maintenance using machine learning for engineering systems through real-time sensor data and anomaly detection models. *International Journal of Research Publication and Reviews*, 2025. URL <https://doi.org/10.55248/gengpi.6.0725.2541>.
- Thies Oelerich, Florian Beck, Christian Hartl-Nesic, and Andreas Kugi. Boundmpc: Cartesian path following with error bounds based on model predictive control in the joint space. *The International Journal of Robotics Research*, 2025. URL <https://doi.org/10.1177/02783649241309354>.
- Wataru Ohata and Jun Tani. Investigation of the sense of agency in social cognition, based on frameworks of predictive coding and active inference: A simulation study on multimodal imitative interaction. *Frontiers in Neurorobotics*, 2020. URL <https://doi.org/10.3389/fnbot.2020.00061>.
- Shengjie Pan and Gang Wang. Perioperative systemic stress phenotypes predict immune recovery failure, molecular residual disease, and long-term outcomes after colorectal cancer surgery: A prospective cohort study. *World Journal of Surgery*, 2026. URL <https://doi.org/10.1002/wjs.70360>.
- Ioannis Papadakis. Structural failure mode analysis of the binary goldbach conjecture. *Mathematics and Computer Science*, 2026. URL <https://doi.org/10.11648/j.mcs.20261102.11>.
- Bhushan Pardeshi. Logistic regression analysis for prediction of financial failure: Evidence from central public sector enterprises in india. *Vision: The Journal of Business Perspective*, 2022. URL <https://doi.org/10.1177/09722629221135241>.

- Roshan Patel, Sophia Colmenares, and Michael A. Webb. Sequence patterning, morphology, and dispersity in single-chain nanoparticles: Insights from simulation and machine learning. *ACS Polymers Au*, 2023. URL <https://doi.org/10.1021/acspolymersau.3c00007>.
- Leonel Patricio, Leonilde Varela, and Zilda de Castro Silveira. Proposal for a sustainable model for integrating robotic process automation and machine learning in failure prediction and operational efficiency in predictive maintenance. *Applied Sciences*, 2025. URL <https://doi.org/10.3390/app15020854>.
- Xiuyan Peng, Bo Wang, Lanyong Zhang, and Peng Su. Shipboard power management for failure mode using the hybrid mpc approach. *Energies*, 2021. URL <https://doi.org/10.3390/en14102915>.
- Yang Peng, Huaicheng Yan, Kai Rao, Penghui Yang, and Yunkai Lv. Distributed model predictive control for unmanned aerial vehicles and vehicle platoon systems: a review. *Intelligence and Robotics*, 2024. URL <https://doi.org/10.20517/ir.2024.19>.
- Bartosz Poskart, Grzegorz Iskierka, Kamil Krot, Robert Burduk, Pawe Gwizdal, and Arkadiusz Gola. Multi-parameter predictive model of mobile robot’s battery discharge for intelligent mission planning in multi-robot systems. *Sensors*, 2022. URL <https://doi.org/10.3390/s22249861>.
- Stefano Primatesa, Abdalla Osman, and Alessandro Rizzo. Mp-rrt#: a model predictive sampling-based motion planning algorithm for unmanned aircraft systems. *Journal of Intelligent and Robotic Systems*, 2021. URL <https://doi.org/10.1007/s10846-021-01501-3>.
- Kathleen M. Quinlan and Edd Pitt. Towards signature assessment and feedback practices: a taxonomy of discipline-specific elements of assessment for learning. *Assessment in Education Principles Policy and Practice*, 2021. URL <https://doi.org/10.1080/0969594x.2021.1930447>.
- Przemyslaw Rajzer and Jan Biegus. Sacubitril/valsartan in a wide spectrum of heart failure patients (from mechanisms of action to outcomes in specific populations). *Heart Failure Reviews*, 2025. URL <https://doi.org/10.1007/s10741-024-10471-1>.
- Roza Ranjbar, Javier G. Martin, Jose M. Maestre, Lucien Etienne, Eric Duviella, and Eduardo F. Camacho. Stochastic model predictive control of an irrigation canal with integrated performance-driven path planning of a measurement robot. *Journal of Hydroinformatics*, 2025. URL <https://doi.org/10.2166/hydro.2025.300>.
- Ahmad Reda, Ahmed Bouzid, and Jozsef Vasarhelyi. Model predictive control for automated vehicle steering. *Acta Polytechnica Hungarica*, 2020. URL <https://doi.org/10.12700/aph.17.7.2020.7.9>.
- Chao Ren, Chunli Li, Liang Hu, Xiaohan Li, and Shugen Ma. Adaptive model predictive control for an omnidirectional mobile robot with friction compensation and incremental input constraints. *Transactions of the Institute of Measurement and Control*, 2022. URL <https://doi.org/10.1177/01423312211021321>.
- Connor Riley, Pascal van Hentenryck, and Enpeng Yuan. Real-time dispatching of large-scale ride-sharing systems: Integrating optimization, machine learning, and model predictive control. *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence*, 2020. URL <https://doi.org/10.24963/ijcai.2020/609>.
- Jonathan Rogers. Stochastic model predictive control for guided projectiles under impact area constraints. *Journal of Dynamic Systems, Measurement, and Control*, 2015. URL <https://doi.org/10.1115/1.4028084>.
- Frank Ruschitzka, Robert Lukowski, Stefano Corda, Christian Meier, and Peter Sandner. Use of soluble guanylyl cyclase stimulators in heart failure therapy-a mode of action perspective. *ESC Heart Failure*, 2026. URL <https://doi.org/10.1093/escf/xvag117>.
- Adam Safron. The radically embodied conscious cybernetic bayesian brain: From free energy to free will and back again. *Entropy*, 2021. URL <https://doi.org/10.3390/e23060783>.

- Md Sadman Sakib and Yu Sun. Star: A foundation model-driven framework for robust task planning and failure recovery in robotic systems. *International Journal of Artificial Intelligence and Robotics Research*, 2025a. URL <https://doi.org/10.1142/s2972335325500073>.
- Md Sadman Sakib and Yu Sun. Star: A foundation model-driven framework for robust task planning and failure recovery in robotic systems. *arXiv*, 2025b. URL <http://arxiv.org/abs/2503.06060v1>.
- Fahed H. Salahat, Hayder A. Rasheed, and Huthaifa I. Ashqar. Ml-ccd: machine learning model to predict concrete cover delamination failure mode in reinforced concrete beams strengthened with frp sheets. *Software Impacts*, 2024. URL <https://doi.org/10.1016/j.simpa.2024.100685>.
- S. M. Bhagya P. Samarakoon, M. A. Viraj J. Muthugala, and Mohan Rajesh Elara. Toward obstacle-specific morphology for a reconfigurable tiling robot. *Journal of Ambient Intelligence and Humanized Computing*, 2021. URL <https://doi.org/10.1007/s12652-021-03342-2>.
- Max Schwenzer, Muzaffer Ay, Thomas Bergs, and Dirk Abel. Review on model predictive control: an engineering perspective. *The International Journal of Advanced Manufacturing Technology*, 2021. URL <https://doi.org/10.1007/s00170-021-07682-3>.
- Carmelo Sferrazza, Michael Muehlebach, and Raffaello D’Andrea. Learning-based parametrized model predictive control for trajectory tracking. *Optimal Control Applications and Methods*, 2020. URL <https://doi.org/10.1002/oca.2656>.
- Srushti B. Shah. Ai-powered pipe failure prediction: Reducing excavation costs by 60% with robotics + ml. *International Journal of Social Sciences, Language and Linguistics*, 2025. URL <https://doi.org/10.55640/ijssll-05-11-01>.
- Amir-Mohammad Shamekhi, Amir Taghavipour, and Amir H. Shamekhi. Engine idle speed control using nonlinear multiparametric model predictive control. *Optimal Control Applications and Methods*, 2020. URL <https://doi.org/10.1002/oca.2581>.
- Rui Shan, Ping Sun, Shuoyu Wang, and Hongbin Chang. Each step time-limited iterative learning control for a cushion robot with motion velocity constraints. *Transactions of the Institute of Measurement and Control*, 2024. URL <https://doi.org/10.1177/01423312231190446>.
- Komal Sharma, Chetan Sharma, Shamneesh Sharma, and Evans Asenso. Broadening the research pathways in smart agriculture: Predictive analysis using semiautomatic information modeling. *Journal of Sensors*, 2022. URL <https://doi.org/10.1155/2022/5442865>.
- Gengjin Shi, Miaomiao Ma, Donghai Li, Yanjun Ding, and Kwang Y. Lee. A process-model-free method for model predictive control via a reference model-based proportional-integral-derivative controller with application to a thermal power plant. *Frontiers in Control Engineering*, 2023. URL <https://doi.org/10.3389/fcteg.2023.1185502>.
- David Hyunchul Shim, H.J. Kim, and Shankar Sastry. Decentralized nonlinear model predictive control of multiple flying robots. *Robotics literature*, 2003. URL <https://doi.org/10.1109/cdc.2003.1271710>.
- Joao Silva, G. Wagner, R. Ataide Da Silva, Antonio Morais, Joao Ribeiro, Sacha Trevelyan Mould, Bruno Figueiredo, J. M. Nobrega, and Paulo J. S. Cruz. Real-time precision in 3d concrete printing: Controlling layer morphology via machine vision and learning algorithms. *Inventions*, 2024. URL <https://doi.org/10.3390/inventions9040080>.
- Priscile Fogou Suawa, Anja Halbinger, Marcel Jongmanns, and Marc Reichenbach. Noise-robust machine learning models for predictive maintenance applications. *IEEE Sensors Journal*, 2023. URL <https://doi.org/10.1109/jsen.2023.3273458>.
- Olivia Sulistina. Reinforcement learning-based model predictive controller for mobile robots in dynamic environments with safety constraints. *Journal of Electrical Engineering*, 2025. URL <https://doi.org/10.54209/elimensi.v3i02.408>.

- Naoki Takahashi and Kenichiro Nonaka. Model predictive leg configuration control for leg/wheel mobile robots that adapts to changes in ground level. *Journal of Robotics and Mechatronics*, 2023. URL <https://doi.org/10.20965/jrm.2023.p0160>.
- E Talen. Success, failure, and conformance: An alternative approach to planning evaluation. *Environment and Planning B: Planning and Design*, 1997. URL <https://doi.org/10.1068/b240573>.
- Marco Tamborini. The material turn in the study of form: From bio-inspired robots to robotics-inspired morphology. *Perspectives on Science*, 2021. URL https://doi.org/10.1162/posc_a_00388.
- Weiqiang Tang, Wenkun Long, and Haiyan Gao. Model predictive control of hypersonic vehicles accommodating constraints. *IET Control Theory and Applications*, 2017. URL <https://doi.org/10.1049/iet-cta.2017.0265>.
- Trym Tengesdal, Tor Arne Johansen, Tom Daniel Grande, and Simon Blindheim. Ship collision avoidance and anti grounding using parallelized cost evaluation in probabilistic scenario-based model predictive control. *IEEE Access*, 2022. URL <https://doi.org/10.1109/access.2022.3215654>.
- Akshay Thirugnanam and Koushil Sreenath. Dynamic incentive selection for hierarchical convex model predictive control. *arXiv*, 2025. URL <http://arxiv.org/abs/2502.04642v1>.
- Karthick Thiagarajan, Sarath Kodagoda, and Linh Nguyen. Predictive analytics for detecting sensor failure using autoregressive integrated moving average model. *Robotics literature*, 2017. URL <https://doi.org/10.1109/iciea.2017.8283153>.
- Nikhil M. Thoppil, V. Vasu, and C. S. P. Rao. Failure mode identification and prioritization using fmeca: A study on computer numerical control lathe for predictive maintenance. *Journal of Failure Analysis and Prevention*, 2019. URL <https://doi.org/10.1007/s11668-019-00717-8>.
- Thomas George Thuruthel, Egidio Falotico, Federico Renda, and Cecilia Laschi. Learning dynamic models for open loop predictive control of soft robotic manipulators. *Bioinspiration and Biomimetics*, 2017. URL <https://doi.org/10.1088/1748-3190/aa839f>.
- Tri Tran, K-V. Ling, and Jan M. Maciejowski. Model predictive control of nonlinear input-affine systems with feasibility and stability constraints. *2014 13th International Conference on Control Automation Robotics and Vision (ICARCV)*, 2014. URL <https://doi.org/10.1109/icarcv.2014.7064441>.
- Mohammed Turkey and Paul M. H. Dummer. Predictive failure mode analysis: The potential role of artificial intelligence in predicting failure modes during root canal retreatment. *Australian Endodontic Journal*, 2026. URL <https://doi.org/10.1111/aej.70036>.
- Jenica N. Upshaw, Marvin A. Konstam, David van Klaveren, Farzad Noubary, Gordon S. Huggins, and David M. Kent. Multistate model to predict heart failure hospitalizations and all-cause mortality in outpatients with heart failure with reduced ejection fraction. *Circulation: Heart Failure*, 2016. URL <https://doi.org/10.1161/circheartfailure.116.003146>.
- Elena Villalba-Aguilera, Joaquim Blesa, and Pere Ponsa. Model-based predictive control for position and orientation tracking in a multilayer architecture for a three-wheeled omnidirectional mobile robot. *Robotics*, 2025. URL <https://doi.org/10.3390/robotics14060072>.
- Hongru Wang and Jinzhi Wang. Distributed model predictive consensus control for linear multiagent system with coupled state constraints. *Systems and Control Letters*, 2026. URL <https://doi.org/10.1016/j.sysconle.2025.106337>.
- Kai Wang and Sebastien Gros. Recursive feasibility of stochastic model predictive control with mission-wide probabilistic constraints. *2021 60th IEEE Conference on Decision and Control (CDC)*, 2021. URL <https://doi.org/10.1109/cdc45484.2021.9683028>.

- Kangbo Wang, Zhengqi Xu, Kaijie Zhang, Yating Huang, and Jun Xu. Lattice piecewise affine approximation of explicit nonlinear model predictive control with application to trajectory tracking of mobile robot. *IET Control Theory and Applications*, 2024a. URL <https://doi.org/10.1049/cth2.12553>.
- Yongchao Wang, Tian Zheng, Maged Iskandar, Marion Leibold, and Jinoh Lee. Practical and robust incremental model predictive control for flexible-joint robots. *Mechatronics*, 2025. URL <https://doi.org/10.1016/j.mechatronics.2025.103364>.
- Zelin Wang, Feng Gao, Yue Zhao, Yunpeng Yin, and Liangyu Wang. Improved a* algorithm and model predictive control- based path planning and tracking framework for hexapod robots. *Industrial Robot: the international journal of robotics research and application*, 2023. URL <https://doi.org/10.1108/ir-01-2022-0028>.
- Zhaodong Wang, Lixue Mei, and Xiaoqun Ma. A novel generalized sliding mode controller for uncertain robot manipulators based on motion constraints. *Mechanical Sciences*, 2024b. URL <https://doi.org/10.5194/ms-15-55-2024>.
- Karl Worthmann, Mohamed W. Mehrez, Mario Zanon, George K. I. Mann, Raymond G. Gosine, and Moritz Diehl. Model predictive control of nonholonomic mobile robots without stabilizing constraints and costs. *IEEE Transactions on Control Systems Technology*, 2015. URL <https://doi.org/10.1109/tcst.2015.2488589>.
- Xiwei Wu, Bing Xiao, Lu Cao, and Haibin Huang. Optimal transport and model predictive control-based simultaneous task assignment and trajectory planning for unmanned system swarm. *Journal of Intelligent and Robotic Systems*, 2024. URL <https://doi.org/10.1007/s10846-024-02060-z>.
- Zhe Wu, David Rincon, Quanquan Gu, and Panagiotis D. Christofides. Statistical machine learning in model predictive control of nonlinear processes. *Mathematics*, 2021. URL <https://doi.org/10.3390/math9161912>.
- Xin Jian Xie, Yu Tian, Jia Huang, Qingquan Luo, and Tianxiang Chen. Surgery without distance: will 5g-based robot-assisted telesurgery redefine modern surgery? *Translational Lung Cancer Research*, 2025. URL <https://doi.org/10.21037/tlcr-2025-16>.
- Liping Xu, Fang Cao, Lian Wang, Weihua Liu, Meizhu Gao, Li Zhang, Fuyuan Hong, and Miao Lin. Machine learning model and nomogram to predict the risk of heart failure hospitalization in peritoneal dialysis patients. *Renal Failure*, 2024. URL <https://doi.org/10.1080/0886022x.2024.2324071>.
- C. YANG, Q. WU, and G. JOYCE. Effects of constraints on bipedal balance control during standing. *International Journal of Humanoid Robotics*, 2007. URL <https://doi.org/10.1142/s0219843607001230>.
- Hongjiu Yang, Mingchao Guo, Yuanqing Xia, and Lei Cheng. Trajectory tracking for wheeled mobile robots via model predictive control with softening constraints. *IET Control Theory and Applications*, 2017. URL <https://doi.org/10.1049/iet-cta.2017.0395>.
- Yu Yang, Hongze Xu, and Xiuming Yao. Disturbance rejection event-triggered robust model predictive control for tracking of constrained uncertain robotic manipulators. *IEEE Transactions on Cybernetics*, 2023. URL <https://doi.org/10.1109/tcyb.2023.3305941>.
- Yuanqing Yang and Baocang Ding. Model predictive control for lpv models with maximal stabilizable model range. *Asian Journal of Control*, 2020. URL <https://doi.org/10.1002/asjc.2070>.
- Chen Yao, Zhijun Li, Haiyi Kong, and Ke Fan. Model predictive tracking control of nonholonomic mobile robots with coupled input constraints and unknown dynamics. *IEEE Transactions on Industrial Informatics*, 2018. URL <https://doi.org/10.1109/tii.2018.2874182>.
- Rioshar Yarveysy, Faisal Khan, and Rouzbeh Abbassi. Data-driven predictive corrosion failure model for maintenance planning of process systems. *Computers and Chemical Engineering*, 2022. URL <https://doi.org/10.1016/j.compchemeng.2021.107612>.

- Kiwon Yeom. Path planning for autonomous driving of mobile robots using deep neural network based model predictive control. *International Journal of Emerging Technology and Advanced Engineering*, 2021. URL https://doi.org/10.46338/ijetae1121_04.
- Xiaoming Yin, Jia Li, Chunyu Pan, Gang Liu, Zhenhua Li, and Song Bai. Development and validation of a predictive model for stone-free failure after extracorporeal shockwave lithotripsy in patients with ureteral stone in a large prospective cohort. *World Journal of Urology*, 2023. URL <https://doi.org/10.1007/s00345-023-04358-3>.
- Haiyong Yu, Yanqiong Jin, Yonghao He, and Wei Sui. Efficient task-specific conditional diffusion policies: Shortcut model acceleration and so(3) optimization. *arXiv*, 2025. URL <http://arxiv.org/abs/2504.09927v1>.
- Shuyou Yu, Matthias Hirche, Yanjun Huang, Hong Chen, and Frank Allgower. Model predictive control for autonomous ground vehicles: a review. *Autonomous Intelligent Systems*, 2021. URL <https://doi.org/10.1007/s43684-021-00005-z>.
- Muiz Mahmud Zaim, Siti Fauziah Toha, Mawada Ahmed Mohamed, Muhammad Abdullah, Md Ataur Rahman, and Masjuki Haji Hassan. Motorcycle adaptive cruise control using model predictive control with integrated sensor fusion. *International Journal of Robotics and Control Systems*, 2026. URL <https://doi.org/10.31763/ijrcs.v5i6.2113>.
- Mahdi Zarghami, Ahmad Fakharian, Amin Ganjali Poudeh, and Aras Adhami-Mirhosseini. Model-based predictive control of wheeled omni-directional robots considering nonlinear dynamical constraints and input delay. *2014 13th International Conference on Control Automation Robotics and Vision (ICARCV)*, 2014. URL <https://doi.org/10.1109/icarcv.2014.7064517>.
- Min Zeng, Yifan Wu, Yiming Li, Rui Yin, Chengqian Lu, Junwen Duan, and Min Li. Lnclocformer: a transformer-based deep learning model for multi-label lncrna subcellular localization prediction by using localization-specific attention mechanism. *Bioinformatics*, 2023. URL <https://doi.org/10.1093/bioinformatics/btad752>.
- Hongtao Zhang, Jialing Yao, and Songmei Tian. Model predictive control for trajectory planning considering constraints on vertical load variation. *Electronics*, 2024a. URL <https://doi.org/10.3390/electronics13081488>.
- Jing Zhang, Jiexiong Ding, Qingzhao Li, Qicheng Ding, Zhong Jiang, Wei Wang, and Li Du. An optimization model of high-speed and high-precision machining based on model predictive control. *Volume 4B: Dynamics, Vibration, and Control*, 2018. URL <https://doi.org/10.1115/imece2018-88326>.
- Jingjun Zhang, Zhili Fang, Zhongqi Zhang, Ruizhen Gao, and Shaobo Zhang. Trajectory tracking control of nonholonomic wheeled mobile robots using model predictive control subjected to lyapunov-based input constraints. *International Journal of Control Automation and Systems*, 2022. URL <https://doi.org/10.1007/s12555-019-0814-x>.
- Shi-Yan Zhang, Xiong-Jian Ma, Xue-Xia Zhu, and Na Cai. Exploring predictive markers for liver failure post-hepatectomy in hepatocellular carcinoma patients. *World Journal of Gastrointestinal Surgery*, 2024b. URL <https://doi.org/10.4240/wjgs.v16.i5.1482>.
- Wei Zhang, Ya-Ni Jin, Chang Sun, Xiao-Feng Zhang, Rui-Qi Li, Qin Yin, Jin-Jun Chen, and Yu-Zheng Zhuge. Development and validation of a predictive model for acute-on-chronic liver failure after transjugular intrahepatic portosystemic shunt. *World Journal of Gastrointestinal Surgery*, 2024c. URL <https://doi.org/10.4240/wjgs.v16.i5.1301>.
- Yifeng Zhang, Daqi Zhu, Mingzhi Chen, and Simon X. Yang. Model predictive tracking control based on adaptive sliding mode constraints for unmanned underwater vehicles. *Intelligence and Robotics*, 2026. URL <https://doi.org/10.20517/ir.2026.06>.
- A. Zheng and M. Morari. Stability of model predictive control with mixed constraints. *IEEE Transactions on Automatic Control*, 1995. URL <https://doi.org/10.1109/9.467664>.

1566 Hao Zhou and Gursel Alc. A magnetically actuated novel robotic capsule for site-specific drug
1567 delivery inside the gastrointestinal tract. *IEEE Transactions on Systems Man and Cybernetics*
1568 *Systems*, 2021. URL <https://doi.org/10.1109/tsmc.2021.3088952>.
1569
1570 Guojin Zhu, Shuyang Yu, Yu Ning, Xuhua Ren, Peng Wei, and Yu Wu. Numerical simulation on the
1571 progressive failure processes of foundation pit excavation based on a new particle failure method.
1572 *Geofluids*, 2021. URL <https://doi.org/10.1155/2021/7374363>.
1573
1574 Tianqi Zhu, Jianliang Mao, Linyan Han, and Chuanlin Zhang. Fuzzy adaptive model predictive control
1575 for image-based visual servoing of robot manipulators with kinematic constraints. *International*
1576 *Journal of Control, Automation and Systems*, 2024. URL <https://doi.org/10.1007/s12555-022-0205-6>.
1577
1578 Zixin Zhuang, Benjamin S. Mashford, Liang Zheng, and T. Daniel Andrews. Towards a cytometry
1579 foundation model: Interpretable sample-level predictive modelling via pretrained transformers.
1580 *Robotics literature*, 2026. URL <https://doi.org/10.64898/2026.03.31.712806>.
1581
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